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Microwave Absorption of Aluminum/Hydrogen Treated Titanium Dioxide Nanoparticles

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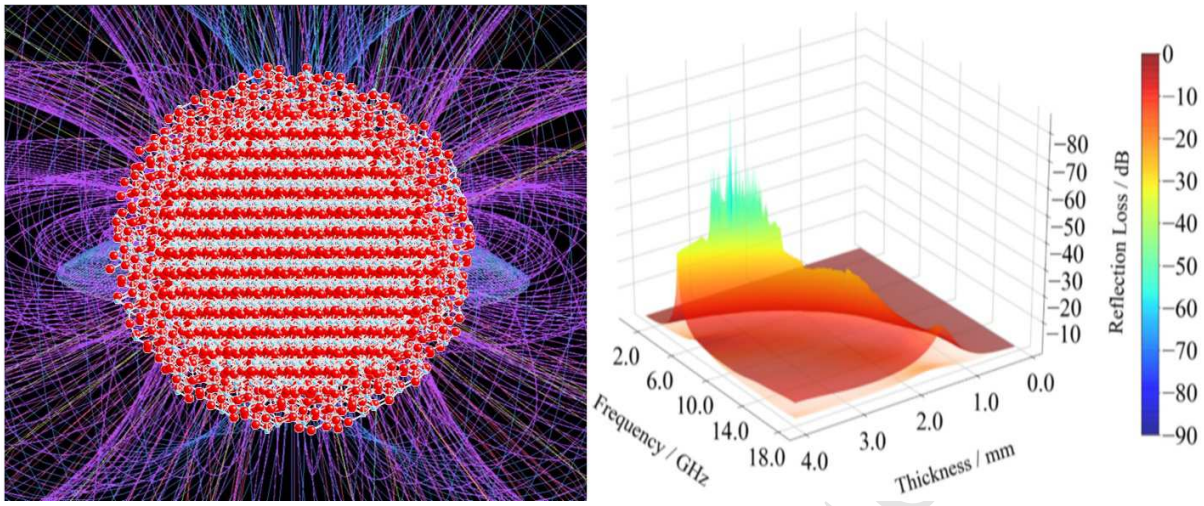
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H_2-Al treated TiO_2 nanoparticles display great microwave absorption performance.

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KEYWORDS: Microwave absorption; black TiO_2 nanoparticles; hydrogenation; aluminum reduction;

ABSTRACT Interactions between incident electromagnetic energy and matter are of critical importance for numerous civil and military applications such as photocatalysis, solar cells, optics, radar detection, communications, information processing and transport, et al. Traditional mechanisms for such interactions in the microwave frequency mainly rely on dipole rotations and magnetic domain resonance. In this study, we present the first report of the microwave absorption of Al/H_2 treated TiO_2 nanoparticles, where the Al/H_2 treatment not only induces structural and optical property changes, but also largely improves the microwave absorption performance of TiO_2 nanoparticles. Moreover, the frequency of the microwave absorption can be finely controlled with the treatment temperature, and the absorption efficiency can reach optimal values with a careful temperature tuning. A large reflection loss of -58.02 dB has been demonstrated with 3.1 mm TiO_2 coating when the treating temperature is 700 °C. The high efficiency of microwave absorption is most likely linked to the disordering-induced property changes in the materials. Along with the increased microwave absorption properties are largely increased visible-light and IR absorptions, and enhanced electrical conductivity and reduced skin-depth, which is likely related to the interfacial defects within the TiO_2 nanoparticles caused by the Al/H_2 treatment.

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1. INTRODUCTION

Understanding electromagnetic radiation and its interactions with matter has been of pertinent importance in our pursuits to develop the next generation of advanced technologies.[1] Specifically, radiation defined by the microwave region of the electromagnetic spectra (1-18 gigahertz) has been utilized for radar detection,[2–4] communications purposes,[1,4–6] information processing and transport,[1,5,6] et alia. For certain systems, integrating materials into various devices as to interact with electromagnetic radiation in specific desirable fashions, like coatings for radar absorption in stealth technology, or absorption of emitted electromagnetic radiation from electronic devices so that information stored therein is secure, are material traits that remain highly desired[1–5,7] As such, discoveries of new materials[8–10] that possess intrinsic capabilities for microwave absorption are vital to the future technology development.

Traditionally, due to a lack of interaction mechanisms between titanium dioxide (TiO_2) and microwave radiation, TiO_2 has historically not been considered an effective microwave absorbing material.[11–13] Conventional models of interaction between microwave photons and materials have been largely confined to interactions between an associated electromagnetic field, and a material that possesses either permanent and/or induces dipoles for dipole rotation, or magnetic moment alignment for ferromagnetic resonance, [4–6,14,15] until recently, where introduction of the new mechanism proposed as the collective-movement-of-interfacial-dipole (CMID) model for hydrogenated TiO_2 nanoparticles provided an alternate approach. In this model, a collective interfacial polarization amplified microwave absorption (CIPAMA) occurs due to the interaction of the incident microwave irradiation and the collective dipoles from the build-up of charge densities at the crystalline/disordered interfacial boundaries within

hydrogenated core/shell TiO_2 nanoparticles.[11] This process of hydrogenation[16] has brought forth multitudes of new methodologies for perturbation of TiO_2 that has been highly studied since its discovered use in water photolysis by Fujishima et al in the early 1970's.[17] Since first being synthesized in 2011, this so-christened "black TiO_2" has generated enormous amounts of interest globally,[18–20] because of both its demonstrated ability to improve the optical properties of the nanocrystal, and the methods potential to be coupled with already proven techniques of material perturbation, such as metal ion doping,[21–24] non-metal doping,[24–27] and self-doping,[24,28–30] to generate a broad array of new materials with enhanced light/matter interactions.[30–32] For black TiO_2, the exposure of an initial TiO_2 grain to elevated temperatures in a hydrogen atmospheric environment induces partial recrystallization from an initial crystalline phase to that of a disordered hydrogenated phase at the nanoscale.[11–13,16,24] This nanoscale partial recrystallization induces engineered disorder at the material surface phase boundary, that can be effectively fine-tuned by both time and temperature exposure variance. The disordering process of crystallization has been shown to perturb the chemical system of TiO_2 so to generate a material that demonstrates intrinsic capabilities for microwave absorption.[11–13,16]

Lattice disordering of TiO_2 nanoparticles has been shown recently to be achieved not only via hydrogenation, but also via utilizing aluminum reduction at elevated temperatures.[33–36] The aluminum processing resulted in similar modifications to the structural, optical, and photocatalytic properties of TiO_2 nanoparticles. From such observations, it seems reasonable to believe that a combination of the aluminum reduction with hydrogenation can be postulated to allow for, at minimum, perturbation, and at best, enhancement, for the microwave absorbing capabilities of TiO_2 nanoparticles. As such, in this study we investigate the structural, optical,

and microwave dielectric, magnetic, and reflection loss properties of TiO_2 nanoparticles treated with aluminum metal and hydrogen gas simultaneously at elevated temperatures. The final analysis of these Al/H_2 treated TiO_2 nanoparticles demonstrate that the material possesses impressive microwave absorption performance. As microwave absorption in terms of reflection loss is dependent on permittivity, permeability, incident frequency, and material thickness,[37,38] and both permittivity and permeability are dependent on incident frequency, through the utilization of interpolation techniques in a manner so to increase the sensitivity of analysis,[39] we show that the theoretical point of maximal reflection loss is even greater than the maximal point derived from the experimental data set, and that one of the materials synthesized herein demonstrates a point of perfect reflection loss. Furthermore, we demonstrate herein that the treating temperature has a large influence on the position and value of the microwave absorption; as the synthesis temperature increases, the frequency of maximal microwave absorption decreases, and the overall magnitude of maximal absorption increases first with temperature, reaches apex, and subsequently decreases. Our study shows that the microwave absorption of the Al/H_2 treated TiO_2 nanoparticles can be fine-tuned simply with controlling the treatment temperature. Meanwhile, the treated TiO_2 nanoparticles possess a structural disordering near the surface of the nanoparticles, a large visible-light absorption, an increased IR absorption, a large Raman luminescent background, a much-improved microwave electrical conductivity, and much reduced skin-depth. This study leads us to conclude that the structural defects and minor chemical composition changes play the underlying key roles in the dramatic changes for various light-matter interactions, based on general understandings of the structure/composition determination.

2. EXPERIMENTAL

Commercial anatase TiO_2 particles and aluminium powders were obtained from Sigma-Aldrich and Fisher Scientific, respectively, and used without further purification. In a typical synthesis, 1.0 g TiO_2 was grinded 0.3 g Al powder for 10 min with a pestle and mortar, then put in a high-temperature tube furnace and treated at certain temperatures (600, 650, 700 and 750°C) under hydrogen atmosphere for 3 hours. The cooled sample was washed with 1.0 M HCl solution overnight under stirring, and subsequently filtered and then dried at 100 °C overnight to obtain the aluminium-hydrogen treated TiO_2 nanoparticles.

The scanning electron microscopy (SEM) images were captured via a Vega3 SEM instrument operating at an electron voltage of 20 kV over 376.9 and 4.75 µm width of field for images S1A and S1B respectively. The transmission electron microscopy (TEM) and high-resolution transmission electron microscopy (HRTEM) images were obtained on a FEI Tecnai F200 TEM instrument with an electron accelerating voltage of 200 kV. A small amount of sample dispersed in water was dropped onto a thin holey carbon film, and dried overnight before the TEM measurement. The X-ray diffraction (XRD) patterns were collected on a Rigaku Miniflex XRD instrument with Cu Kα as the X-ray source (wavelength = 1.5418 Å). Rietveld refinement was executed via Profex software utilizing the included anatase structure file for initialization,[40] and the resultant was visualized using the VESTA 3 program.[41] The surface chemical bonding information was obtained with XPS on a PHI 5400 XPS system equipped with a conventional (non-monochromatic) Al anode X-ray source with K_α radiation. A small amount of TiO_2 nanocrystals was pressed onto conductive carbon tape for XPS measurements. The binding energies were calibrated with respect to the C 1s peak from the carbon tape at 284.6 eV. The Raman spectra were measured on an EZRaman-N benchtop Raman spectrometer with the excitation wavelength of 785 nm. The Fourier transform infrared (FTIR) spectra were collected

using a Thermo-Nicolet iS10 FT-IR spectrometer with an attenuated total reflectance (ATR) unit where TiO_2 nanoparticles were pressed onto its ZnSe window directly and the measurement were conducted in air at room temperature. The ultraviolet-visible (UV-vis) spectra were collected with an Agilent Cary 60 UV-Vis spectrometer with a fiber optical reflectance unit. MgO power was used as the reference material. The microwave complex permittivity and permeability of anatase and hydrogenated TiO_2 nanosheets were measured at the frequency range of 1 – 18 GHz using HP8722ES network analyser. The samples dispersed in melted paraffin wax were pressed into a ring with the thickness of 2.0 mm, inner diameter of 3 mm, and outer diameter of 7 mm. The mass content of sample was controlled at 60 wt%, and the testing was performed at room temperature.

3. RESULTS AND DISCUSSION

The bulk material and crystal lattice properties of the synthesized materials can be observed directly with SEM, TEM, and HRTEM imaging. **Figure S1** shows the SEM images of the Al/H_2 treated TiO_2 nanoparticles, demonstrating the fine nature of the materials over the wide field of view shown, and general uniform elemental composition as evidenced by the backscattering electron resultant being generally uniform in contrast against the carbon-based platform. **Figure 1** shows the TEM and HRTEM images of the Al/H_2-TiO_2 nanoparticles, the selected images also coming from material treated at 700°C. The TEM images in **Figure 1A** and **Figure 1B** indicate the primary particle size is around 70-100 nm. The HRTEM images in **Figure 1C** and **Figure 1D** show the generation of a 1-2 nm thick disordered layer near the surface of the nanoparticles with highly crystalline lattice fringes in the bulk (as can be also seen from the selected area electron diffraction (SAED) in Figure 1A-1B and fast Fourier transform

patterns in Figure 1C-1D in the insets). As such, these Al/H_2 treated TiO_2 nanoparticles possess the deviation in structural manifestation referred to in the literature as a core/shell nanostructure, in that the core of the crystalline structure remains unperturbed, and perturbation of the crystalline material is induced primarily along the surface. This observation is consistent with previous findings on both hydrogenated TiO_2 nanoparticles and aluminum-reduced TiO_2 nanoparticles;[15,32-35] the Al/H_2 treatment brings a similar structural modification on TiO_2 nanoparticles as both hydrogenation and aluminum reduction bring individually.

The bulk crystal structure of the TiO_2 nanoparticles treated at 600, 650, 700 and 750 °C, along with the starting commercial anatase TiO_2 nanoparticles, was analyzed via XRD and is shown in **Figure 2A**. The primacy of the crystalline anatase phase was shown to be maintained between over the entirety of the material treatment range. However, demonstrated at the 750 °C treatment temperature is the formation of the reduced magneli phase heptaoxotetratitanate, Ti_4O_7, COD ID 1008048, which is not demonstrated in the previous studies of hydrogenation or aluminum reduction, where thermal conversion of TiO_2 occurs from the anatase to rutile phases upon the subsequent increase of the temperature, as the rutile phase is the more stable phase at higher temperatures (e.g. > 700 °C), even in pure hydrogen environment.[11] Herein however, due to the introduction of aluminum material into the hydrogenation system, a reduced phase of Ti_4O_7 is observed, which is a novel observation compared to previous reports on both hydrogenated TiO_2 and Al-reduced TiO_2 particles.[11–13,33–36] Such seems to suggest that the aluminum additive is both having a reducing effect on the system and is synergistically compatible with the hydrogenation process for means of material reduction and restructuring. The nanoparticle sizes were calculated using the Scherrer equation, $\tau = K\lambda/(\beta \cos\theta)$, where τ is the mean size of the crystallite domains, K is the shape factor with a typical value of 0.9, λ is the

X-Ray wavelength, β is the line broadening full width at half maximum (FHWM) peak height in radians, and θ is the Bragg angle.[15,42] Using the FHWM values from the (200) peak, the crystallite sizes are 37.2, 36.9, 35.5, 35.7, and 44.1 nm for anatase, and the Al/H_2-TiO_2 Material treated at 600, 650, 700 and 750 °C, respectively. Therefore, the nanoparticle size of the anatase phase is slightly decreased below 700 °C and largely increased at 750 °C, which coincides with the manifestation of the Ti_4O_7 phase (**Figure 2A**, demarcated via 'o' symbolism). The following diffraction curves were furthermore analyzed via Rietveld Refinement so to elucidate the change in the anatase domain as a function of increasing treatment temperature. The results of such analyses are presented in both **Figure S2** and **Figure S3** in regard to calculated diffraction and lattice parameters for the anatase phase as the material undergoes thermal treatment. The unit cell volume of the anatase crystalline domain was calculated to be 136.21 $Å^3$ for starting anatase, and 136.03 $Å^3$, 136.17 $Å^3$, 136.12 $Å^3$, and 136.65 $Å^3$ for the 600, 650, 700, and 750 °C Al/H_2-TiO_2 nanomaterials. Such suggests that the thermal treatment both at and below 700 °C had a sort of pressurizing effect on the core anatase domain of the Al/H_2 treated nanomaterials, inducing compression of the grain. However, at 750 °C the stressors present within the processing system perturb the nanoparticle in a reverse fashion, resulting in growth of the resultant particles and expansion of the crystalline domain along with the introduction of the Ti_4O_7 magneli phase. This crystal deviation, coupled with the understanding that reduction processes typically force TiO_2 crystal domains from a corner-shared octahedral manifestation to that of an edge-shared octahedral, or from an edge-shared octahedral, to that of a plane-shared octahedral, could furthermore have effects on the bulk dielectric and magnetic properties of the generated nanomaterials.[43]

Figure 2B shows the Raman spectra of the nanomaterials of interest. The signal resultant of the anatase nanomaterial is a typical characteristic spectrum for the material, with no background and strong peaks at 161, 414, 535, and 660 cm^{-1} from the Ti–O bond stretching modes A_{1g} at 513 cm^{-1}, B_{1g} at 519 cm^{-1}, and E_g at 660 cm^{-1}, as well as the O–Ti–O bending modes B_{1g} at 414 cm^{-1}, and E_g at 164 cm^{-1} in the crystalline anatase phase.[7,16,44] While maintaining these characteristic peaks, the treated TiO_2 nanoparticles have some extent of broad luminescence backgrounds resultant from surface defects, consistent with previous studies of hydrogenated TiO_2 nanoparticles.[11–13] The integral volume of luminescence background increases in the following order: anatase < 650 °C < 700 °C < 750 °C < 600 °C. Furthermore, the position of the luminescence background locates mainly in the lower wavenumber range for the TiO_2 treated at 600 °C, but higher in wavenumber range for the TiO_2 treated at 750 °C. Such would seem to suggest that there are differing manifestations/distributions in material defects with respect to the various treated TiO_2 nanoparticles.

The FTIR spectra are shown in **Figure 2C** for the transmittance percentage (%T) of incident infrared radiation over the range of 4000-650 cm^{-1}. Anatase TiO_2 nanoparticles showed a transmittance of nearly 100% from 4000 to 1300 cm^{-1}, which gradually decreases to 52.3% at 650 cm^{-1} due to lattice vibrations. Meanwhile, a miniscule divot is observed at appx. 1640 cm^{-1}, and a shallow, prolonged depression in %T between 3800 and 3000 cm^{-1}, centered around 3300 cm^{-1}, are observed in the response pattern. Both are related to O-H vibrational modes – the former from surface hydroxyl groups, the latter from chemisorbed water.[45] The overall transmittance of the treated TiO_2 nanoparticles decreases as the synthesis temperature is iteratively increased. The hydroxyl group markers near 3300 and 1640 cm^{-1} decrease in prominence due to the treatment. The transmittance starts to gradually decrease from 2200 to 650

cm^{-1} for the TiO_2 nanoparticles treated at 600 and 650 °C, where the magnitude of decrease becomes larger as the wavenumber becomes smaller. The transmittance starts to decrease from 2800 to 650 cm^{-1} for the TiO_2 nanoparticles treated at 700 °C, with a considerable decrease in the lower wavenumber region. Distinct decreases in the transmittance are observed for the TiO_2 nanoparticles treated at 750 °C. The transmittance is only 80.3% at 4000 cm^{-1}, and gradually decreases to 37.5% at 650 cm^{-1}. The region below 860 cm^{-1} shows a sharp decrease in transmittance from the lattice Ti-O bond vibrations in the TiO_2 nanoparticles.[45–47] The decrease in the transmittance from 4000 to 860 cm^{-1} correlates with the large luminescence background in the Raman spectra, again, being likely related to the defects created in the TiO_2 nanoparticles after the Al/H_2 treatment process. The trend of the transmittance decrease against temperature increase is well aligned with the increase of the Raman background, with an exception for the TiO_2 nanoparticles treated at 600 °C, where the discrepancy can most likely be attributed to the different selecting rules for the corresponding optical transitions involved with the defects in the infrared and Raman spectrum.[48] The large decrease of the transmittance, or the increase of absorption of the infrared irradiation, is often related to the possible free charge carriers existing in the Al/H_2 treated TiO_2 nanoparticles, whereby thermal treatment of the TiO_2 nanomaterials yields materials which due to increased oxygen vacancies in the particle domain demonstrate a lowered Fermi level for the nanoparticles, thus allowing for less hindered charge excitation into the conduction band [45,49–51]. Such understanding is consistent with the previous observations that hydrogenated TiO_2 nanoparticles have much larger electrical conductivity and carrier densities compared to that of anatase TiO_2 nanoparticles.[11,49–55]

Figure 2D displays the UV-vis absorption spectra. The anatase TiO_2 nanomaterial demonstrates significant absorption in the UV region below the wavelength of 380 nanometers,

and subsequently barely any absorption in the visible-light range, which is consistent with the original studies of photocatalysis and the anatase bandgap. The Al/H_2-TiO_2 nanomaterials display considerable absorption characteristics over the entire visible-light region, in addition to the characteristic anatase absorption in the UV region. The quantity of light absorbed in the visible-light region increases in the order of 650 °C < 600 °C < 700 °C < 750 °C. Compared to the moderate increase in the visible-light absorption of the samples treated below 700 °C, the samples treated at 750 °C displays a distinct increase. This increase might be related to the newly formed magneli phase, as TiO_2 interfaces can demonstrate long-wavelength absorption performances across phase boundaries, for example in anatase-rutile composites.[56] As such, the Al/H_2 TiO_2 nanomaterials are shown to display large absorption in the visible-light region, consistent with the Raman and FTIR observations previously mentioned, due to the creation of structural defects and the disordered layer on the outside of the crystalline TiO_2 nanoparticles.[11–13,33–36]

Material analysis via XPS was used to investigate the elemental composition and chemical perturbation specifically within the disordered phase, as XPS is a surface-probe technique that optimizes near the material surface at depths of 1-2 nm,[32,57–59] which is consistent with the thickness of the disordered layer in the Al/H_2 treated TiO_2 nanoparticles. **Figure 3** shows the XPS spectra of the TiO_2 nanoparticles treated at 700 °C, mainly showing the O, Ti and C signals. The carbon signal is from the atmospheric deposition during the XPS measurement and is used as the reference with the C 1s core-level binding energy of 284.6 eV. The O 1s core-level XPS spectrum was deconvoluted into a dominant peak near 529.7 eV from the lattice O^{2-} ions in the TiO_2 nanoparticles, and a minor peak near 532.0 eV, which is of a binding energy similar to that of typical hydroxyl groups. The Ti 2p core-level XPS spectrum as

shown in **Figure 3C** demonstrates the characteristic peaks of Ti^{4+} ion of TiO_2 nanoparticles, with peaks centered at 458.7 and 464.3 eV for the Ti $2p_{1/2}$ and Ti $2p_{3/2}$ binding energies, respectively;[11–13,16] if Ti^{3+} ions were generated in the material shell domain, especially in enough quantity so to be the prominent mechanism for microwave absorption, their manifestation in XPS results would be shown as a peak around 457.1 eV,[60] which is not demonstrated. Finally, the Al 2p core-level XPS spectrum revealed a small symmetric signal peak around 75.8 eV, shown in **Figure 3D**, where the symmetry suggests that such signal is the result of non-metallic aluminum particulates within the material domain, such as aluminum oxide or hydroxide,[57] though these crystal phases are not observed in the bulk-phase XRD pattern.

The relative complex permittivity and permeability, $\varepsilon_r = (\varepsilon` - j\cdot\varepsilon``)$ and $\mu_r = (\mu` - j\cdot\mu``)$, were measured over the microwave range from 1-18 GHz. Permittivity and permeability are representations of a materials' intrinsic dielectric and magnetic response to incident electromagnetic radiation, made relative by taking the ratio of the material response with respect to that of the permittivity and permeability of free space, $\varepsilon_0 = 8.854\cdot10^{-12}$ F/m and $\varepsilon_0 = 4\pi\cdot10^{-7}$ H/m respectively. The real portions of these relative values, $\varepsilon`$ and $\mu`$ respectively, represent lossless interaction between the respective material and electromagnetic wave, relative to that of vacuum space[11–13] Subsequently, the complex portions of permittivity and permeability, $\varepsilon``$ and $\mu``$ respectively, represent relative lossy interaction between the material and incident wave.[11–13] Finally, the ratio of $x``/x` = tg\delta_x$, where x is either ε or μ, correlates to the dielectric and magnetic dissipation factors respectively, also known as dielectric or magnetic tangential loss, which is the ratio of the relative power dissipated by the material as thermal energy, with respect to the power stored within the material medium.[11–13] This ratio can be

thought of as a sort of net efficiency parameter for the overall energy transfer process, as the product of lossless interaction and the loss tangent describes the power dissipation per unit volume at a given frequency.[61] Further points of interest in the permittivity and permeability resultants are instances where the lossy action of the material as described by the imaginary portion of the complex values maximizes. These instances are referred to as relaxation peaks, and are associated with the resonance of a material, where the energy dissipation rate from stored energy to thermal heat is maximized. The aforementioned loss tangents typically peak at frequencies higher than that of the relaxation peaks, as the net dissipation efficiency over the entire transfer process and the energy transfer rate from lossless interaction to lossy are both dependent on the magnitude of the lossless interaction, which varies as a function of frequency and in accordance to the Kramers-Kronig dispersion relations.[61,62]

As shown in **Figure 4A**, the anatase TiO_2 nanoparticles demonstrate stable lossless dielectric action over the 1-18GHz testing range, maintaining ε` values between 6.1 and 6.5. The $TiO_2$ nanoparticles treated at 600 °C have similar ε` values, except over the range of 15-18 GHz, where the relative lossless interaction is of slightly lesser magnitude. The TiO_2 nanoparticles treated at 650 °C demonstrate a ε` value dropping from 8.1 to 7.7 over the 1.0-13.0 GHz frequency range, from where ε` then descends to 6.1 at 18.0 GHz. The ε` value at 700 °C starts high at 18.8 times the permittivity of free space at 1.0 GHz, from where the magnitude of lossless interaction gradually decreases to 12.0 at 13.1 GHz, to 7.8 at 16.0 GHz, where it stabilizes until 18.0 GHz. Finally, the ε` value of the 750 °C material starts at 28.7 times the permittivity of free space at 1.0 GHz and gradually falls to 27.2 at 4.3 GHz, from where it then increases to 29.5 at 10.1 GHz, plummets to 13.5 at 13.3 GHz, from where it ramps back up to 21.7 at 17.0 GHz, and decreases finally to 21.0 at 18.0 GHz. These results, specifically within

the 10.1 – 17.0 GHz frequency range across the four material are characteristic of dielectric resonance,[61] which strengthens as a function of increasing treatment temperature, or, as the anatase material is perturbed farther and farther from a pristine state. Furthermore, materials analysis demonstrates large overall increases in the ε` values, observed for the $TiO_2$ nanoparticles treated at 700 and 750 °C. Since the lossless interaction described by ε` is indication of the stored electrical energy within the material, therefore, the Al/H_2 treatment at 700 and 750 °C distinctly increase the capability of TiO_2 nanoparticles to store electrical energy. Such affinity for action may be related to the appearance of Ti_4O_7 phase. This increase, analogous to visible-light absorption, may be traced back to the increase of microwave states or transition cross-sections that could be created by the defects located in the disordered layer or in the crystalline/disordered interface.

The lossy action of both anatase and the Al/H_2-TiO_2 nanoparticles as described by ε`` is presented in **Figure 4B**. Anatase nanomaterials are essentially lossless, with a recorded ε`` value between 0.03 and 0.3. For the 600 °C Al/H_2-TiO_2 nanoparticles, the light/matter interaction was also essentially lossless, with the exception of a gradual increase from 0.26 at 12.1GHz to 1.4 at 15.1 GHz, followed with a gradual relax in the ε`` value to 0.83 at 18.0 GHz. The ε`` value for the TiO_2 nanoparticles treated at 650 °C is between 0.90 and 1.40 over the majority of the tested frequency range, with the exception of a dielectric resonance peak with a magnitude of 2.45 centered at 14.0 GHz covering from 11.3 to 16.7 GHz. The extent of the ε`` value peak associated with relaxation in the 700 °C material is located at a similar frequency to that of the peak demonstrated by the 650 °C nanomaterial, though the relaxation peak is of roughly twice the magnitude. The mean interaction of this material is between 4.2 and 5.5 times the lossy permittivity of free space, and the material furthermore demonstrates a relaxation peak with a

relative value of 6.6, centered at 15.1 GHz, covering a range of 10.8 to 17.8 GHz. Finally, the ε`` value for the 750 °C nanomaterial increases from 1.6 at 1.0 GHz to 8.1 at 18.0 GHz, and has three consecutive relaxation peaks represented within the data, with maximal values for lossy action equal to 18.9 centered at 12.1 GHz, covering from 5.2 to 14.2 GHz, 8.7 centered at 15.7 GHz covering from 14.2 to 16.8 GHz, and 8.4 centered at 17.6 GHz covering from 16.8 to 18.0 GHz. The ε`` values of the Al/$H_2$ treated $TiO_2$ nanoparticles are much stronger than that of the anatase $TiO_2$ nanoparticles. As lossy permittivity so described by ε`` is an expression of the dissipation of electrical energies and is furthermore related to the absorption process, the increased ε`` values for the set of Al/H_2 treated TiO_2 nanoparticles therefore suggests that the dissipation of electromagnetic energy through electronic interaction is improved compared to that of the anatase TiO_2 nanoparticles.

From the values of lossless and lossy dielectric action, the dielectric loss tangent can thus be calculated; such results are shown in **Figure 4C**. As the attenuating lossy interaction of the anatase nanomaterial was small, the subsequent $tg\delta_\varepsilon$ value of anatase TiO_2 nanoparticles was also such, resulting in tangential loss values between 0.02 and 0.04 over the frequency range of 1.0-18.0 GHz. The 600°C material demonstrates a gradual increase in $tg\delta_\varepsilon$, from 0.01 at 1.0 GHz to 0.05 at 12.0 GHz, and then furthermore to 0.21 at 15.2 GHz at a quasi-exponential rate, from where it subsequently peaks and then decreases to 0.14 at 18.0 GHz. The $tg\delta_\varepsilon$ value at 650 °C gradually increases from 0.14 at 1.0 GHz to 0.17 at 11.1 GHz, and furthermore to 0.35 at 14.1 GHz in a similar quasi-exponential fashion, from where the loss tangent crests and subsequently decreases to 0.21 at 18.0 GHz. The $tg\delta_\varepsilon$ value for the 700 °C nanomaterial increases gradually from 0.28 at 1.0 GHz to 0.54 at 18.0 GHz, and has a pronounced peak with the value of 0.73 at 15.9 GHz covering from 11.2 to 17.7 GHz. Finally, the $tg\delta_\varepsilon$ value for the 750 °C Al/H_2 TiO_2

nanomaterial increases gradually from 0.05 at 1 GHz to 0.39 at 18.0 GHz, and has three pronounced peaks with a value of 1.01 at 12.6 GHz covering from 5.1 to 14.0 GHz, a value of 0.46 at 15.3 GHz covering from 14 to 16.8 GHz, and a value of 0.40 at 17.6 GHz covering from 16.8 to 18.0 GHz. Compared to that of anatase TiO_2 nanoparticles, the $tg\delta_\varepsilon$ values of Al/H_2 treated TiO_2 nanoparticles are much larger, indicating a much higher net dissipation efficiency of electrical energy. As demarcated in **Figure 4C**, it appears that the $tg\delta_\varepsilon$ peaks may be put into four distinct groups based on their frequency positions, possibly related to four energy dissipation paths inside the treated TiO_2 nanoparticles – the casual nature of which still requires further investigation.

As shown in **Figure 5A**, the magnetic interaction between incident electromagnetic radiation and anatase TiO_2 nanoparticles is minimal, with the lossless interaction as quantified by $\mu`$ was shown to be confined between 0.95 and 1.04 over the frequency range of 1-18 GHz. Contrast to anatase, the $Al/H_2$ treated $TiO_2$ nanoparticles generally have a larger $\mu`$ values. The $\mu`$ values treated between 600 and 750 °C fluctuate between 1.00-1.13 in most frequency regions; however, distinct variations are observed over the frequency range of 12.1-17.3 GHz. For example, the $\mu`$ value at 600 °C increases from 1.07 at 13.0 GHz to 1.18 at 15.5 GHz, and then decreases to 0.87 at 17.3 GHz; the $\mu`$ value at 650 °C decreases from 1.10 at 13.0 GHz to 0.95 at 16.5 GHz; the $\mu`$ value at 700 °C increases from 1.00 at 13.0 GHz to 1.18 at 16.5 GHz, and then decreases to 1.14 at 18.0 GHz; finally, the $\mu`$ value at 750 °C increases from 0.99 at 10.5 GHz to 1.73 at 13.3 GHz, then decreases to 1.08 at 16.0 GHz and finally increases to 1.18 near 18.0 GHz. The significant difference in this lossless magnetic interaction between the

anatase and Al/H_2 treated TiO_2 nanoparticles suggests that the Al/H_2 treatment also brings in a notable disturbance regarding the stored magnetic energy by the TiO_2 nanoparticles.

In terms of the lossy magnetic interaction as shown in **Figure 5B**, the μ`` value of the anatase $TiO_2$ nanoparticles decreases gradually from 0.30 at 1.0 GHz to 0.04 at 18.0 GHz. In comparison, the μ`` values of the Al/H_2-TiO_2 nanoparticles are smaller than that of the anatase TiO_2 nanoparticles, mainly in the lower frequency range (1.0-11.0 GHz), but are larger in the higher frequency range. For example, the μ`` value of the 600 °C Al/H_2 treated TiO_2 nanoparticles is between 0.01 to 0.1 in the frequency range of 1.0-12.0 GHz, but it increases to 0.30 at 16.4 GHz, and decreases to 0.20 at 18.0 GHz; for the 650 °C nanomaterial, lossy permeability increases to 0.15 at 15.0 GHz and decreases to 0.05 at 18.0 GHz; for the 700 °C nanomaterial, it increases continuously to 0.14 at 18.0 GHz; finally for the 750 °C nanomaterial, μ`` increases to 0.47 at 12.3 GHz, decreases to 0.40 at 13.0 GHz, then increases more so to 0.57 at 14.3 GHz and decreases to 0.35 at 18.0 GHz. The large difference of μ`` values at 750 °C may be potentially related to the appearance of Ti_4O_7 magneli phase. These deviations in interaction to magnetic stimulus as defined by the increase in the lossy permeability values of the anatase after Al/H_2 treatment suggests that such Al/H_2 treatment disturbs the magnetic energy dissipation properties in the TiO_2 nanoparticles, and apparently, the treating temperature plays an important role on the determination of how the lossy interaction manifests.

As with permittivity, from the experimental permeability values, the magnetic loss tangent as defined as the ratio between the lossy magnetic interaction with respect to the lossless can be thus derived. These changes in such $tg\delta_\mu$ parameter as a function of frequency appear similar to that of the μ`` value, shown in **Figure 5C**. The $tg\delta_\mu$ value of anatase TiO_2

nanoparticles decreases gradually from 0.31 at 1.0 GHz to 0.04 at 18.0 GHz; the $tg\delta_\mu$ value of the Al/H_2 treated TiO_2 nanoparticles is between 0.01 to 0.1 in the frequency range of 1.0-12.0 GHz, but then demonstrates distinct peaks in the spectra: for the 600 °C material, a peak of 0.31 at 16.6 GHz; for the 650 °C material, a peak of 0.15 at 15.2 GHz; for the 700 °C material, two peak values of 0.11 at 15.0 and 18.0 GHz; and for the 750 °C material, two peaks of 0.38 at 12.0 GHz and 0.48 at 15.2 GHz. These apparently increased $tg\delta_\mu$ values at higher frequencies suggests that the magnetic dissipation efficiency is largely improved, mainly in the high frequency ranges for the TiO_2 nanoparticles after Al/H_2 treatment.

Figure 6A shows the electrical conductivity (σ) in the frequency range of 1-18 GHz. The σ value is calculated via $\sigma(S{\cdot}m^{-1}) = 2\pi f(\varepsilon_0{\cdot}\varepsilon``)$, where $\varepsilon_0$ is the permittivity of free space equal to $8.854 \times 10^{-12}$ F/m, *f* is the incident frequency in Hertz, and ε`` is the relative imaginary component of permittivity. [2,3,63] Anatase TiO_2 nanoparticles demonstrate minimal conductivity that increases from 0.016 $S{\cdot}m^{-1}$ at 1.0 GHz to 0.068 $S{\cdot}m^{-1}$ at 18.0 GHz, with three small broad peaks apexed around 0.057 $S{\cdot}m^{-1}$ at 6.3 GHz, 0.089 $S{\cdot}m^{-1}$ at 13.4 GHz, and 0.14 $S{\cdot}m^{-1}$ at 17.8 GHz. The σ value is slightly increased in the 600 °C Al/H_2-TiO_2 nanomaterial, especially at higher frequency ranges. Such conductivity increases from 0.0047 $S{\cdot}m^{-1}$ at 1.0 GHz to 0.82 $S{\cdot}m^{-1}$ at 18.0 GHz, and has three weak peaks of 0.026 $S{\cdot}m^{-1}$ at 4.0 GHz, 0.10 $S{\cdot}m^{-1}$ at 9.0 GHz, 1.10 $S{\cdot}m^{-1}$ at 15.2 GHz and one strong peak of 0.91 $S{\cdot}m^{-1}$ at 16.4 GHz. For the 650 °C material, conductivity increases from 0.067 $S{\cdot}m^{-1}$ at 1.0 GHz to 1.35 $S{\cdot}m^{-1}$ at 18.0 GHz, and demonstrates two weak peaks of 0.24 $S{\cdot}m^{-1}$ at 4.0 GHz and 0.62 $S{\cdot}m^{-1}$ at 9.1 GHz, and one strong peak of 1.91 $S{\cdot}m^{-1}$ at 14.1 GHz. The electrical conductivity is further increased after treated at 700 °C; the 700 °C nanomaterial increases from 0.31 $S{\cdot}m^{-1}$ at 1.0 GHz to 4.1 $S{\cdot}m^{-1}$ at 18.0 GHz, and has two weak

peaks of 1.07 $S{\cdot}m^{-1}$ at 3.8 GHz and 2.36 $S{\cdot}m^{-1}$ at 9.0 GHz and furthermore one strong peak of 5.59 $S{\cdot}m^{-1}$ at 15.3 GHz. Finally, the 750 °C Al/H_2-TiO_2 nanomaterial demonstrates the greatest increase in electrical conductivity; such σ value increases from 0.098 $S{\cdot}m^{-1}$ at 1.0 GHz to 6.8 $S{\cdot}m^{-1}$ at 18.0 GHz, and has three peaks of 12.6 $S{\cdot}m^{-1}$ at 12.1 GHz, 7.6 $S{\cdot}m^{-1}$ at 15.8 GHz and finally 8.2 $S{\cdot}m^{-1}$ at 17.7 GHz. The trend in conductivity as a function of synthesis temperature increases in the following order: anatase TiO_2 << 600 °C < 650 °C < 700 °C < 750 °C, and the Al/H_2 treatment largely increases conductivity of the TiO_2 nanoparticles in the majority of the tested microwave range. For example, the conductivity value of 12.6 $S{\cdot}m^{-1}$ at 12.1 GHz for the TiO_2 nanoparticles treated at 750 °C is approximately 140 times of that of the anatase TiO_2 nanoparticles (0.089 $S{\cdot}m^{-1}$ at 13.4 GHz).

Figure 6B shows the skin-depth (δ) of microwave irradiation on the Al/H_2-TiO_2 nanoparticles. The skin-depth is defined as the distance over which the electromagnetic field intensity attenuates by a factor of e^{-1} as it traverses a given material medium, whereby a typical exponential decay function $e^{-\alpha x}$ yields at a distance x = e^{-1} the value equivalent to the inverse of the attenuation constant.[63] In terms of permittivity and permeability, this skin depth can be calculated from the expression $(\delta / mm) = 10^3{\cdot}\alpha^{-1} \approx 10^3{\cdot}(\pi f \mu_0 \mu_r \sigma)^{-1/2}$; herein 10^3 is the conversion factor so to yield results in units mm^{-1}, α is the attenuation constant calculated via the equation $\alpha = j(2\pi f{\cdot}c^{-1}){\cdot}Re\{sqrt[(\mu` - j{\cdot}\mu``)(\varepsilon` - j{\cdot}\varepsilon``)]\}$, j is the complex value of the $sqrt(-1)$, f is the incident frequency in Hertz, c is the speed of light, and both $\varepsilon_r = (\varepsilon` - j{\cdot}\varepsilon``)$ and $\mu_r = (\mu` - j{\cdot}\mu``)$ are the respective relative permittivity and permeability values for lossless and lossy action, respectively, and finally σ is the conductivity of the material, whose functional relation to the skin depth is commonly utilized for means of calculation, particularly in instances where $\varepsilon`` >>$

ε`.[3,63,64] As seen from **Figure 6B**, the δ value decreases as the frequency of the electromagnetic field increases in the microwave region of 1–18GHz. The δ value of anatase TiO_2 nanoparticles starts around 96.3 mm at 1.0 GHz and gradually decreases to 37.5 mm at 7.0 GHz, from where it remains relatively stable over the remainder of the frequency range. The δ value of the Al/H_2-TiO_2 nanoparticles treated at 600 °C generally decays over the 1-18 GHz frequency domain, though it demonstrates localized points of maximal electromagnetic penetration of 241.1 mm at 5.4 GHz and 85.4 mm at 10.3 GHz. For the Al/H_2-TiO_2 nanoparticles treated at 650 °C, the trend in conductivity was similar to that of the 600 °C nanomaterial but at half the length and without the local peaks. A similar observation with the 700 °C nanomaterial, with the overall trend being half the magnitude of the 650 °C material. Finally, the 750 °C Al/H_2-TiO_2 nanoparticles demonstrated a skin-depth values between that of the 650 and 700 °C material beneath the frequency of 9GHz, possibly due to the introduction of the magneli phase. In general, the lower the microwave electrical conductivity, the higher is the ability of the electromagnetic field to propagate and penetrate within the material. As the frequency increases, the δ value decreases. These calculations show that δ value decreases after treatment at temperatures at or higher than 650 °C, and that the electromagnetic field is strongly attenuated; consequently, the electromagnetic field decreases more rapidly.

From the experimentally derived permittivity and permeability vales, it can be shown that the Al/H_2 treatment on TiO_2 nanoparticles increases the dissipation efficiency of not only the electrical component, but also the magnetic component of an incident electromagnetic field, which may lead to the increase of the overall absorption of microwave radiation. As such, the

reflection loss (RL) as a function of frequency and material thickness was calculated via the following equations:

$$RL(db) = 20\log_{10}\left[\left|\frac{Z_{in}-1}{Z_{in}+1}\right|\right] \quad (1)$$

$$Z_{in} = \left[\frac{\mu` - j\cdot\mu``}{\varepsilon` - j\cdot\varepsilon``}\right]^{\frac{1}{2}} \cdot \tanh\left(j\frac{2\pi f\cdot d}{c}\left[(\varepsilon` - j\cdot\varepsilon``)(\mu` - j\cdot\mu``)\right]^{\frac{1}{2}}\right) \quad (2)$$

Here, reflection loss as represented by eqn (1) is a function of the normalized input impedance Z_{in}, represented by eqn (2), which is itself a function of complex relative permittivity and permeability $\varepsilon_r = (\varepsilon` - j\cdot\varepsilon``)$ and $\mu_r = (\mu` - j\cdot\mu``)$, the incident frequency of electromagnetic energy *f*, and the material thickness *d*.[37,65–67] Finally, *j* is the complex value of the *sqrt(-1)*, and *c* is the speed of light. The resultant domains of these calculations are presented in **Figure 7**, over a 1-18GHz, 0.1-4 mm domain, and with a collapsed thickness axis in **Figure 8A**, as well as a collapsed material axis in **Figure 8B**. For example, at a 2.0 mm thickness, the anatase TiO_2 nanoparticles demonstrated a minimal RL value of -2.57 dB at 14.47 GHz. Comparatively, the Al/H_2-TiO_2 nanoparticles demonstrated much larger RL values. For example, minimal RL values of -18.35 dB at 16.81 GHz, -33.70 dB at 14.43 GHz, -25.34 dB at 10.61 GHz and -16.35 dB at 6.95 GHz were reported for the TiO_2 nanoparticles treated at 600, 650, 700 and 750 °C, respectively (**Figures 7A-D**). Globally, the minimal RL values for each of the materials is -4.65 dB at 7.12 GHz, 4.0mm for anatase, -22.77 dB at 16.64 GHz, 1.9 mm for the 600 °C material, -33.7 dB at 14.43 GHz, 2.0 mm for the 650 °C material, -58.02 dB at 6.61 GHz, 3.1 mm for the 700 °C material, and finally -28.26 dB at 8.14 GHz, 1.7 mm for the 750 °C material, again demonstrating the superiority of the Al/H_2-TiO_2 nanoparticles. In terms of the quantification of microwave absorption, the smaller the RL value, the less the microwave reflection from the

material domain and thus the larger the microwave absorption. For example, a RL value of -20 dB corresponds to an absorption of 99.0%, and -30 dB is 99.9% absorption. From the experimental results, the global absorption efficiency of the Al/H_2 treated TiO_2 nanoparticles increases in the following order with respect to the treatment temperature: 600 °C < 650 °C < 750 °C < 700 °C. Deviation in this trend for the 750 °C material could be from the introduction of the more conductive Ti_4O_7 phase within the material medium. Furthermore, the frequency of the absorption shifts to lower values as the treating temperature increases; this trend appears universal over all represented thicknesses and materials, as shown in **Figure 8B**. Such observation demonstrates that the materials presented herein possess the capacity to be tuned via simply changing the treating temperature, so that the generated material interacts with a particular frequency of interest at a particular thickness of interest, ultimately leading to a higher domain of utility for the Al/H_2-TiO_2 nanomaterials.

Using the function relationship between incident frequency and material thickness as previously reported, the 700 °C and 750 °C Al/H_2 TiO_2 nanomaterials were analyzed along the prominent RL band so to observe the reflection loss response as a function of simulated permittivity and permeability values. In order to accomplish such, the $\chi_m = \mu_r / \mu_0 - 1$ magnetic susceptibility parameter was arbitrarily set to zero, so to simulate the complex relative permeability as the trivial result of $\mu_r = (1 - j \cdot 0)$.[15] Furthermore, simulations of material interaction where there was no electrical dissipation, i.e. where $\varepsilon`` = 0$ for the incident electromagnetic wave, were executed so to observe a change in function response.[68] The comparative results are presented in **Figure S4** and **Figure S5**, where S4 is the 700 °C material, and S5 is the 750 °C material, respectively. As can be seen from the resultant deviations in the reflection loss curves, removing the contribution of the magnetic component from the 700 °C

material had little to no effect on the overall reflection loss value in the areas in which the experimental results demonstrate a strong response for reflection loss. Contrastingly, removing the dielectric contribution from the reflection loss function drastically effects the resultant data set, to the point where there is essentially no reflection loss within the general vicinity of the maximal RL value of the experimental dataset, and little RL response over the entirety of the RL band. This is consistent with the previously proposed CMID model,[11] which suggests that a resultant electric field across the phase boundary of a core/shell nanoparticle is the mechanism of action which is inducing reflection loss. Interestingly, for the 750 °C material, although the point of maximal reflection loss appears to also be contributed to completely by the dielectric response, the removal of the lossy dielectric action from the calculation shows that the given magnetic parameters from the material would induce a strong response in reflection loss, if there was indeed no dielectric action (**Figure S5**). However, as there is dielectric action, the resultant permittivity/permeability coupling negates experimental validation of this magnetic pathway, as the experimental data demonstrates only a weak RL response at the equivalent frequency/thickness coordinates. This may be due to the introduction of the Ti_4O_7 magneli phase, though the nature of this interaction requires further investigation. Overall, the modeling process previously described seems to validate the CMID model, in that the results of simulated responses suggest that removing that dielectric light/matter interaction as defined by the permittivity of the nanomaterial results in the negating of the reflection loss value we see from both the 700 °C and 750 °C Al/H_2-TiO_2 nanomaterial.

It should be noted that, as the measurement of materials analysis for complex permittivity and permeability generates a discrete set of permittivity and permeability values, which are then iteratively computed across the frequency-thickness domain to determine reflection loss, the

completeness of a given data set is highly sensitive to the interval step size over which the experimental procedure is executed. As the process of deriving reflection loss curves from permittivity and permeability data sets is highly sensitive,[65] it is entirely possible that the analysis would not or could not have been conducted at a small enough frequency/thickness interval, and in terms of points of perfect reflection loss, where the normalized input impedance was exactly equal to the value of 1,[69] and thus the point which achieved such impedance value from a material data set was glossed over unintentionally. This leaves researchers in the field of microwave absorbing materials to chance, as it is extremely difficult to accurately predict the necessary six input values for permittivity, permeability, frequency, and thickness *a priori* before running the final calculation for reflection loss. This issue can be mitigated however through further theoretical analysis via the implementation of interpolation techniques;[70] as permittivity and permeability are a function of frequency, interpolation allows for the reflection loss value to be a true function instead of an iterated calculation, though the functionalization of the normalized input impedance:

$$Z_{in}(f,d) = \left[\frac{\mu`(f) - j\cdot\mu``(f)}{\varepsilon`(f) - j\cdot\varepsilon``(f)}\right]^{\frac{1}{2}} \cdot \tanh\left(j\frac{2\pi f\cdot d}{c}\{[\varepsilon`(f) - j\cdot\varepsilon``(f)][\mu`(f) - j\cdot\mu``(f)]\}^{\frac{1}{2}}\right) \quad (3)$$

Herein, the normalized input impedance as represented by equation 3 is strictly a function of frequency and thickness, as the permittivity and permeability values are replaced with permittivity and permeability functions, defined by an interpolation function via the independent input frequency parameter. This input impedance is then used in equation 1 as a substitute for equation 2 so to calculate reflection loss. Both linear and cubic spline interpolating techniques were used; linear interpolation simply assumes a straight line between two successive points in a

data set, and cubic spline interpolation, which is the most widely utilized of interpolating functions in physics and engineering, utilizes third-order polynomials so to conjoin subsets of data in a manner that ensures continuity in the first and second derivatives.[70] Utilizing a 0.01 GHz·0.01 mm grid spacing in analysis, the maximal reflection loss from the Al/H_2-TiO_2 nanomaterial treated at 750 °C increases to a reported theoretical value of -47.77 dB at 8.30 GHz, 1.66 mm via linear interpolation, and -43.04 dB at 8.30 GHz, 1.66 mm via the cubic spline function (**Figure S6** and **Figure S7**). Furthermore, the reflection loss trend as a function of increased precision was shown to be generally linear (as shown in **Figure S8**) in both the cases of the linear interpolation and cubic spline, which suggests that as precision increases reflection loss goes to negative infinity. From such, it can be said that in this case the Al/H_2-TiO_2 750 °C nanomaterial satisfies the criterion for an optimal absorption point.[37,65] However, these results also demonstrate that the sensitivity of a given material setup increases as the target reflection loss value increases, or in another word, the thickness of a given material is allowed to vary by smaller and smaller distances so to target ranges of smaller and smaller frequencies as the predetermined acceptable threshold for reflection loss increases. As such, it may be that isolated points of perfect reflection loss exist in a given material, which are interesting to consider, but these isolated points become less useful the more one reduces the RL threshold.

Utilizing the cubic spline interpolation function, the effective bandwidth for various RL thresholds was investigated in the Al/H_2-TiO_2 nanomaterial treated at 700 °C across a range of material thicknesses. The effective bandwidths from interpolated data at RL thresholds of -10, -20, -30, and -40dB over the 4.0mm range of material thicknesses is demonstrated in **Figure 9B**, with the maximal achieved effective bandwidths being reported as 6.13GHz with a 1.61mm thickness for the -10 dB RL threshold, 0.84 GHz with a 1.91 mm for the -20 dB RL threshold,

0.21 GHz with a 2.86 mm for the -30 dB RL threshold, and finally 0.09 GHz with a 2.79 mm for the -40 dB RL threshold. This process of interpolation allows for an increased precision in analysis of the effective bandwidth, as shown in **Figure 9B**, where the resultant curve is more precisely defined compared to strictly using the experimental data set. As can be seen in **Figure 9C**, the power function which describes the relationship between the frequency of maximal reflection loss and the material thickness is essentially identical when derived from the experimental vs. the interpolated data set. Here, the associated data sets were found to follow the non-linear regression functions of $f_1(d) = 22.259/d^{1.072}$ for the experimental, and $f_2(d) = 22.201/d^{1.070}$ for the interpolated. These results are both in general agreement with the quarter-wavelength relationship reported in previous works,[9,68] where $d(f)=4^{-1}(2m-1)\cdot c / [\, n(f)\cdot f\,] = \lambda\cdot(2m-1)/4$, with d being the material thickness, f being the frequency, n being the refractive index (which is dependent on permittivity and permeability, and thus ultimately dependent on frequency), c being the speed of light, λ being the propagation wavelength, and m being an integer value greater than zero, where in this specific case $m = 1$. The major deviation in the reported results is in the reported maximal reflection loss values, as shown in **Figure 9D**, and the resultant grid spacing utilized in mapping response over 3D space, as shown in **Figure 9A** and **Figure S10**.

Based on the above results, the Al/H_2 treated TiO_2 nanoparticles display distinct dielectric properties that diverge significantly from anatase. The Al/H_2-TiO_2 nanomaterials possess perturbed crystalline domains, crystalline/disordered core/shell structures, and demonstrate large visible-light absorption, broad IR absorption, and strong microwave absorption, similar to those of hydrogenated or aluminum reduced TiO_2 nanoparticles. Property changes are mainly observed in the TiO_2 nanoparticles treated above 700 °C. The perturbation

process at 750 °C leads to the appearance of the Ti_4O_7 magneli phase, a dramatic visible-light absorption, a large and broad IR absorption, distinct changes of the microwave complex permittivity and permeability, and microwave absorption. Based on the Raman spectra, the quantity of defects qualitatively determined from the luminescence background in the lower wavenumber range tends towards the following order: anatase TiO_2, 650 °C < 700 °C << 750 °C << 600 °C, but tends to follows a different order in the higher wavenumber range: anatase TiO_2, 650 °C, 700 °C < 600 °C < < 750 °C. Based on the UV-visible absorption spectra, the quantity of defects related to the visible-light absorption tends to follow the order of anatase TiO_2 < 650 °C < 600 °C < 700 °C < <750 °C. Based on the FTIR absorption spectra, the quantity of defects related to the broad IR absorption from 650 cm^{-1} to 4000 cm^{-1} follows the given order: anatase TiO_2 < 600 °C, 650 °C < 700 °C < <750 °C. It would appear that these given trends as related to the Raman luminescence background, Visible-light absorption, and IR absorption do not correlate well with each other. Given such, it's reasonable to conclude that the defects responsible for one of the optical transitions may not be active for the other, likely manifesting as different mechanisms for different optical transitions, due to differing crystalline properties which result from the adjusting of the material treating temperature. However, the above defects are likely related to the dielectric properties of the TiO_2 nanoparticles, where the influence of the magnetic properties is minimal, as the oscillatory nature of the electromagnetic field is too fast for the magnetic component to respond in those frequency ranges. The real and imaginary permittivity values tend towards the following order: anatase TiO_2 < 600 °C << 650 °C < 700 °C < 750 °C. Therefore, the change of the relative permittivity values increases with the increase of the treating temperature. However, the dielectric loss tangent ($tg\delta_\varepsilon$) value follows a differing sequence: anatase TiO_2 < 600 °C < 650 °C < 750 °C < 700 °C. The change of the relative

permeability values tends towards the following order: anatase TiO_2 < 700 °C < 650 °C < 600 °C < 750 °C, with the magnetic loss tangent ($tg\delta_\mu$) following suite. From permittivity and permeability, the electrical conductivity and skin-depth was calculated. The overall electrical conductivity follows the order: anatase TiO_2 << 600 °C < 650 °C < 700 °C < 750 °C, and the magnitude of the skin-depth tends towards a differing order: 600 °C > anatase TiO_2 > 650 °C > 750 °C > 700 °C. The change of the microwave efficiency as defined by permittivity and permeability apparently lead to the change of the skin-depth value, and thus the attenuation of electromagnetic radiation as it traverses through the TiO_2 nanoparticles. Ultimately, using the permittivity and permeability parameters in tandem with incident frequency values over a range of material thicknesses, the values for reflection loss were derived; the data shows that the overall microwave absorption efficiency tends towards the following order: anatase TiO_2 << 650 °C < 600 °C < 750 °C < 700 °C. Given the relationship between reflection loss and that of complex permittivity and permeability, the microwave absorption efficiency appears to be controlled by the co-play of the four complex function inputs, as well as the incident electromagnetic frequency and material thickness.[45,71,72] The resultant from removing the lossy action of the material as defined by permittivity suggests that it is electronic interactions that induces the majority of electromagnetic interaction, most likely through interaction with the core/shell phenomena via the CMID model as previously proposed.[11–13] As the trends in microwave absorption against the temperature do not match well with the changes of the visible-light, IR absorption, the Raman luminescence background, or the electrical conductivity, it becomes non-trivial to predict the performance of the microwave absorption based on these individual analytical techniques, likely again, due to the different absorption mechanisms for each event. However, it can be seen that the anatase TiO_2 nanoparticles, which do not have

defects involved for visible-light, IR absorption, the Raman luminescence background or poor electrical conductivity have a poor microwave absorption performance.[11–13] It would seem reasonable from such to conclude that the defects created through the Al/H_2 thermal treatment play the casual role in establishing the strong improvement in microwave absorption performance.[11–13,73,74] As seen from the microscopic images, the treated TiO_2 nanoparticles have crystalline/disordered core/shell structures, which is characteristic to the features defined by the previously proposed CMID model so to account for a materials' enhanced microwave absorption performance. In this study, we observe experimentally that the global minimal RL values for each of the materials is -4.65 dB at 7.12 GHz, 4.0mm for anatase, -22.77 dB at 16.64 GHz, 1.9 mm for the 600 °C material, -33.7 dB at 14.43 GHz, 2.0 mm for the 650 °C material, -58.02 dB at 6.61 GHz, 3.1 mm for the 700 °C material, and finally -28.26 dB at 8.14 GHz, 1.7 mm for the 750 °C material. These results demonstrate the versatility of the synthesized materials, whereby it may be possible to target select frequencies of incident electromagnetic radiation by varying the synthesis temperature, if it is so required that material thickness be held constant. Furthermore, the effective bandwidth of the most promising material, the 700 °C Al/H_2-TiO_2 nanomaterial, at the RL threshold of -10 dB was calculated to be 6.12 GHz from a 1.60 mm thick absorber. Finally, though the utilization of mathematical interpolation techniques we can predict a point of perfect absorption in the 750 °C nanomaterial.

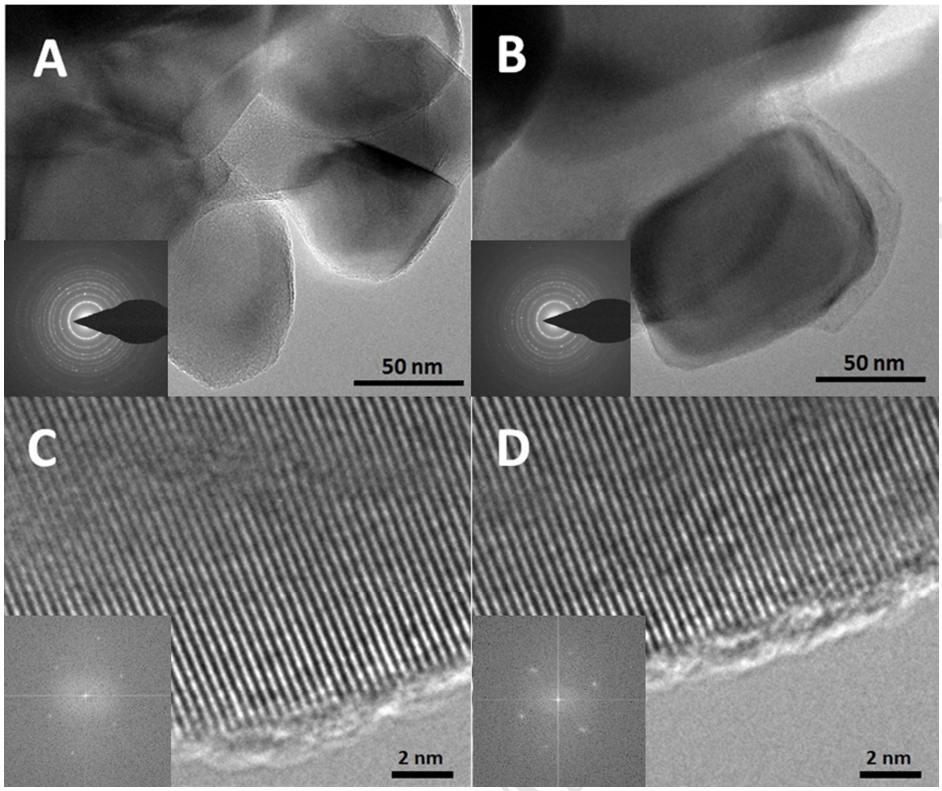
4. CONCLUSIONS

In summary, we have shown in this study that Al/H_2 treated TiO_2 nanoparticles demonstrate intriguing microwave dielectric and absorption properties. With the increase of the treating temperature, the position of the microwave absorption shifts to the lower frequency

region, while there may exist an optimal temperature to reach the maximum microwave absorption efficiency, as the microwave absorption efficiency first increases and then decreases. Accompanied with the changes in the microwave properties are the changes of the visible-light absorption, IR absorption, Raman luminescence background, electrical conductivity, skin-depth, and crystalline structures which are related to various defects and the tiny aluminum doping that are induced by the Al/H_2 treatment and responsible for the microwave absorption performance of the treated TiO_2 nanoparticles. A large RL value of -58.02 dB is achieved with 700 °C treated TiO_2 nanoparticles for a 3.1 mm coating. This action appears to possess fine-tuning capacity, whereby adjusting the treating temperature of the material allows for specific targeting of select frequencies for strong reflection loss. Band analysis of the strongest microwave absorption band suggests that the interaction between the nanoparticle and incident electromagnetic radiation was predominately dielectric in nature. Overall, the high efficiency of microwave absorption is most likely causal linked to the engineered disorder within the nanomaterials, with the resultant reflection loss being induced by dielectric action between the nanomaterial and incident electromagnetic wave.

FIGURES

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685 **Figure 1.** (A, B) Representative TEM and (C, D) HRTEM images of Al/H_2 treated TiO_2 nanoparticles treated at 700 °C. The inserts in (A) and (B) show the SAED patterns and the inserts in (C) and (D) show the FFT images.

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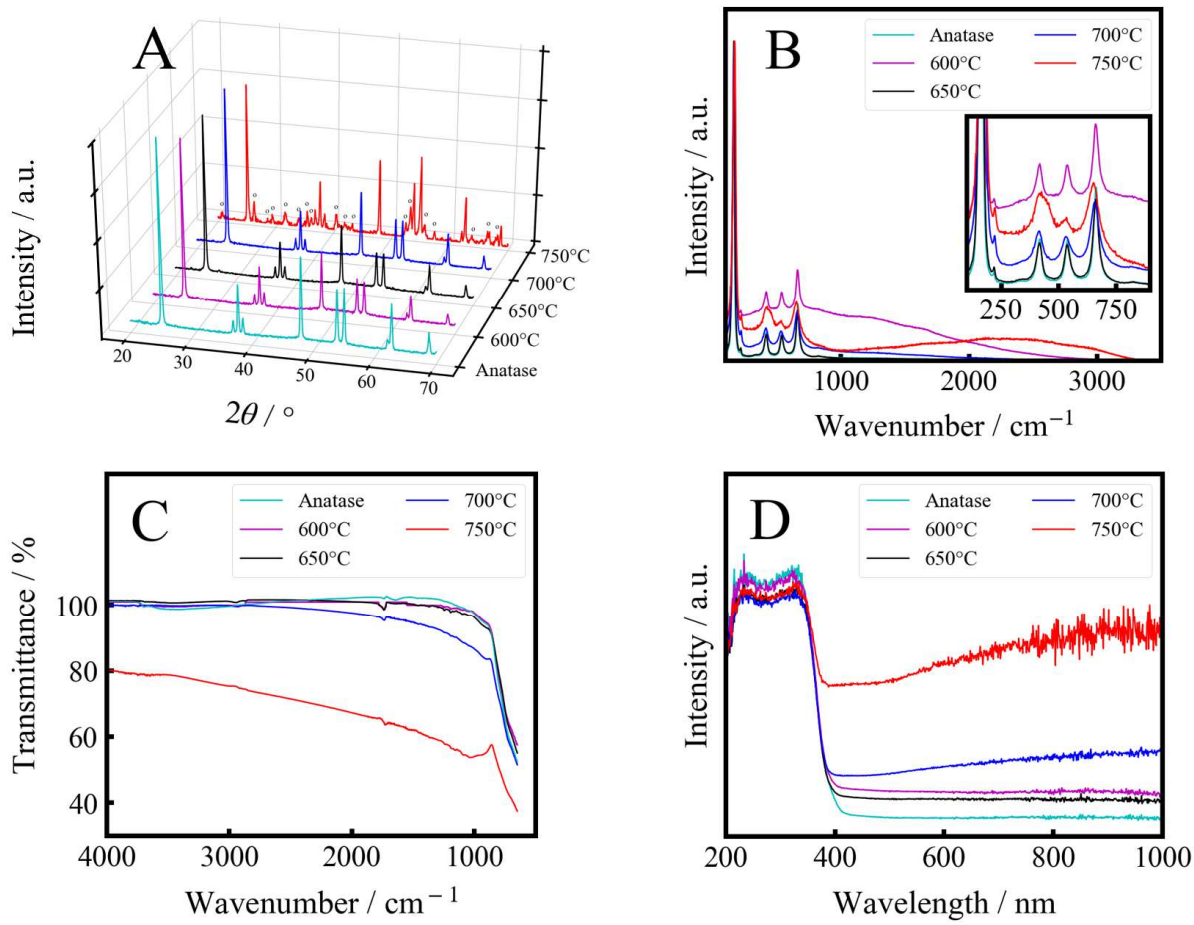
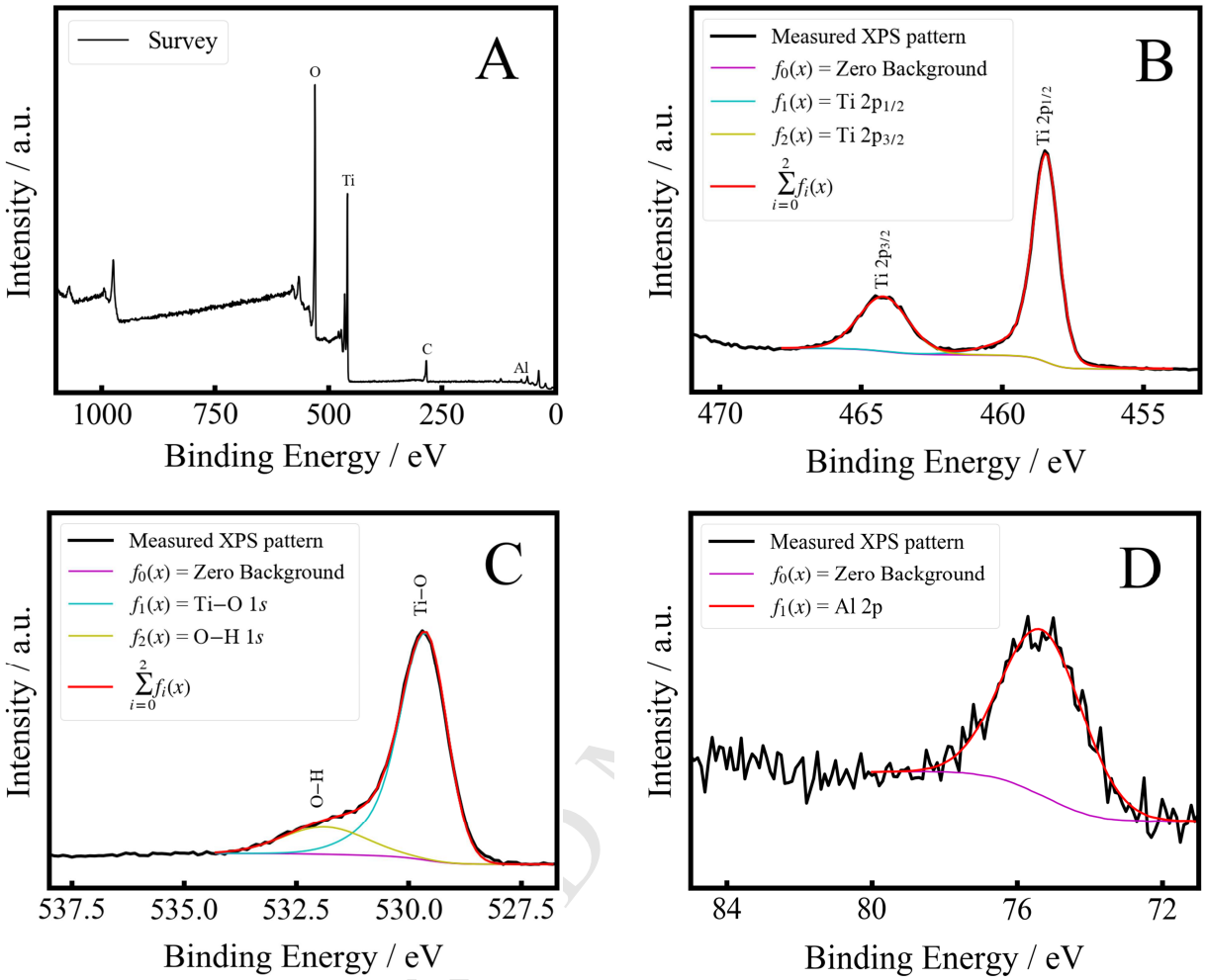
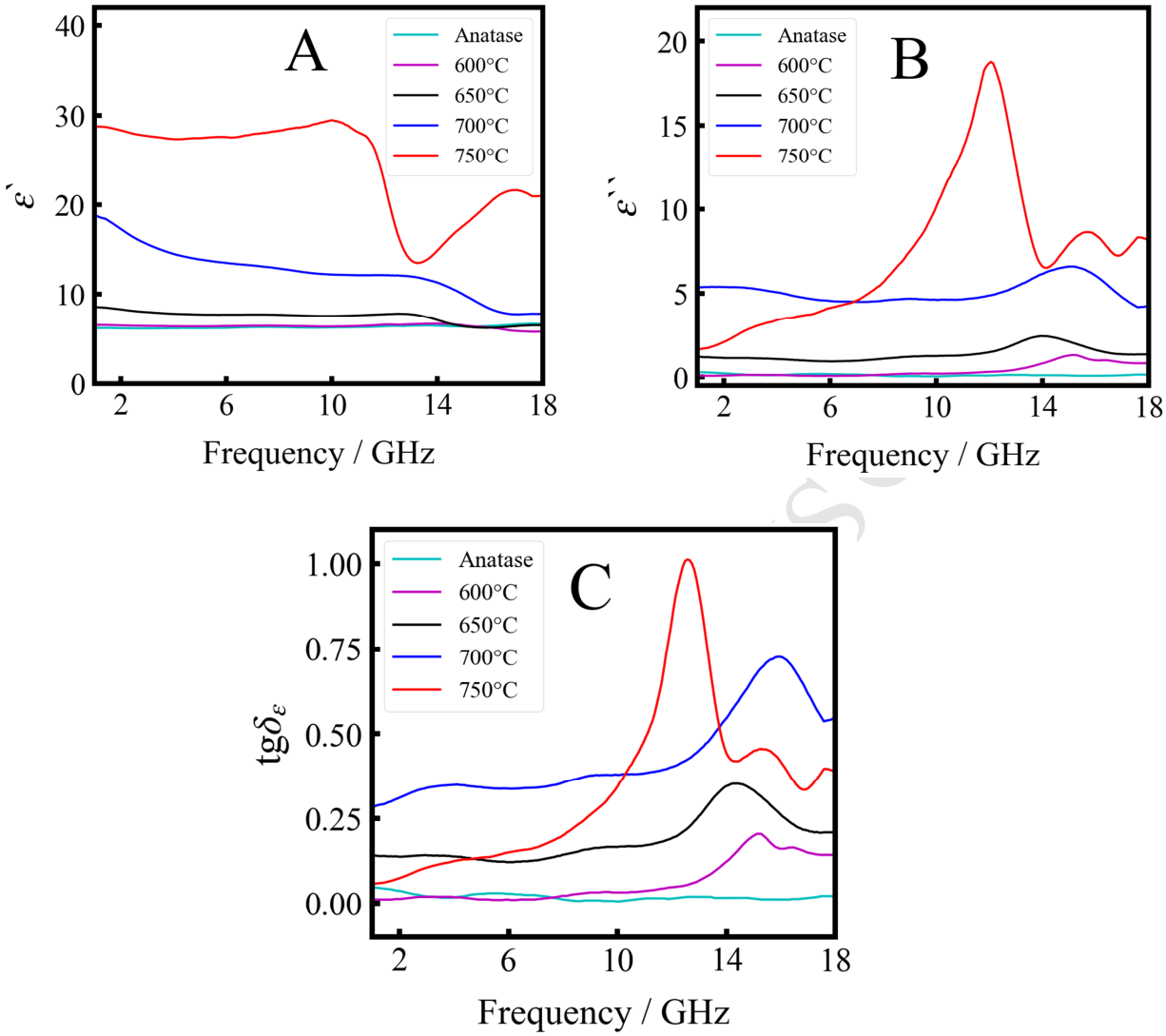


Figure 2. XRD spectra of anatase and TiO_2 nanoparticles treated at 600, 650, 700, and 750 °C. The 'o' demarcation in the Figure designates signal responses due to the presence of Ti_4O_7. (B) Raman, (C) FTIR, and (D) UV-Vis spectra of anatase and the TiO_2 nanoparticles treated at 600, 650, 700, and 750 °C.



700

Figure 3. (A) Survey, (B) Ti 2p, (C) O 1s, and (D) Al 2p core-level XPS spectra of Al/H_2-TiO_2 nanoparticles treated at 700 °C.



705

Figure 5. The frequency profiles of (A) lossless permittivity, (B) lossy permittivity, and (C) the dielectric loss tangent of complex permittivity, over the microwave region of 1–18 GHz, for both anatase and the Al/H_2-TiO_2 nanoparticles treated at 600, 650, 700, and 750 °C.

710

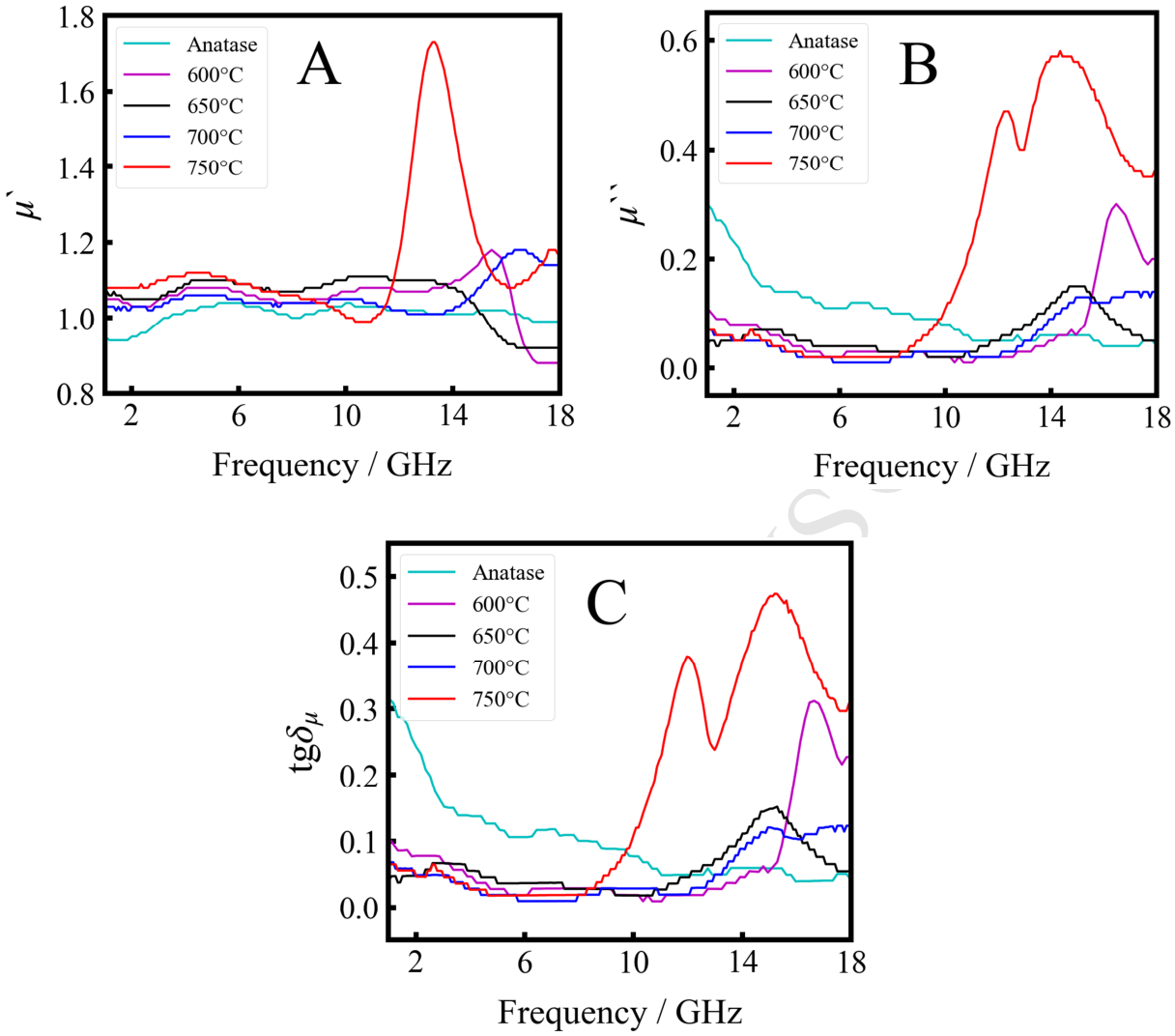
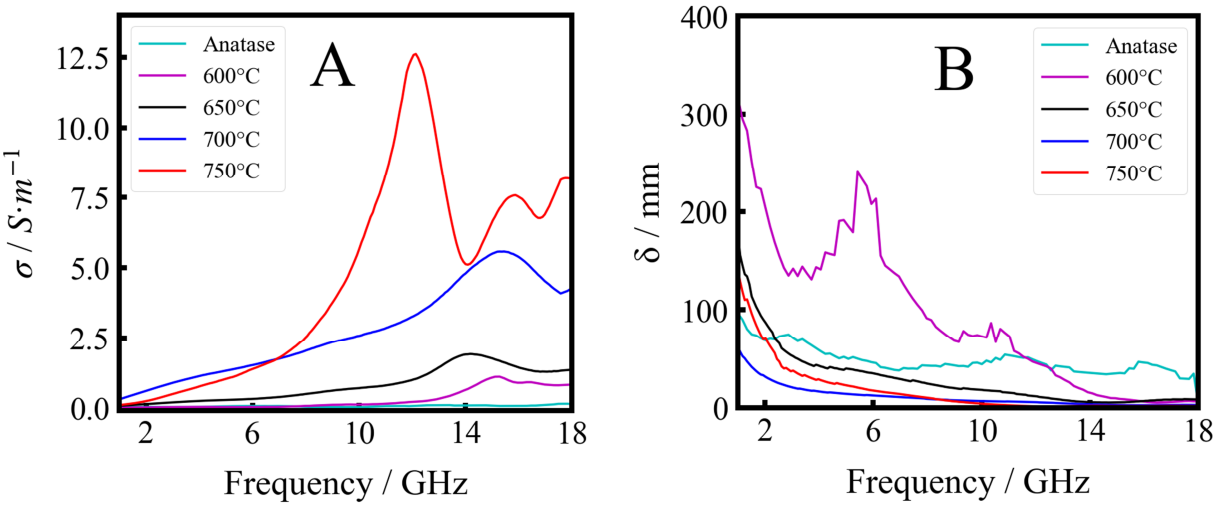


Figure 6. The frequency profiles of (A) lossless permeability, (B) lossy permeability, and (C) the magnetic loss tangent of the complex permeability in the microwave region of 1–18 GHz of anatase and Al/H_2-TiO_2 nanoparticles treated at 600, 650, 700, and 750 °C.



720

Figure 7. (A) Electrical conductivity and (B) skin-depth of anatase and the Al/H_2-TiO_2 nanoparticles treated at 600, 650, 700, and 750 °C.

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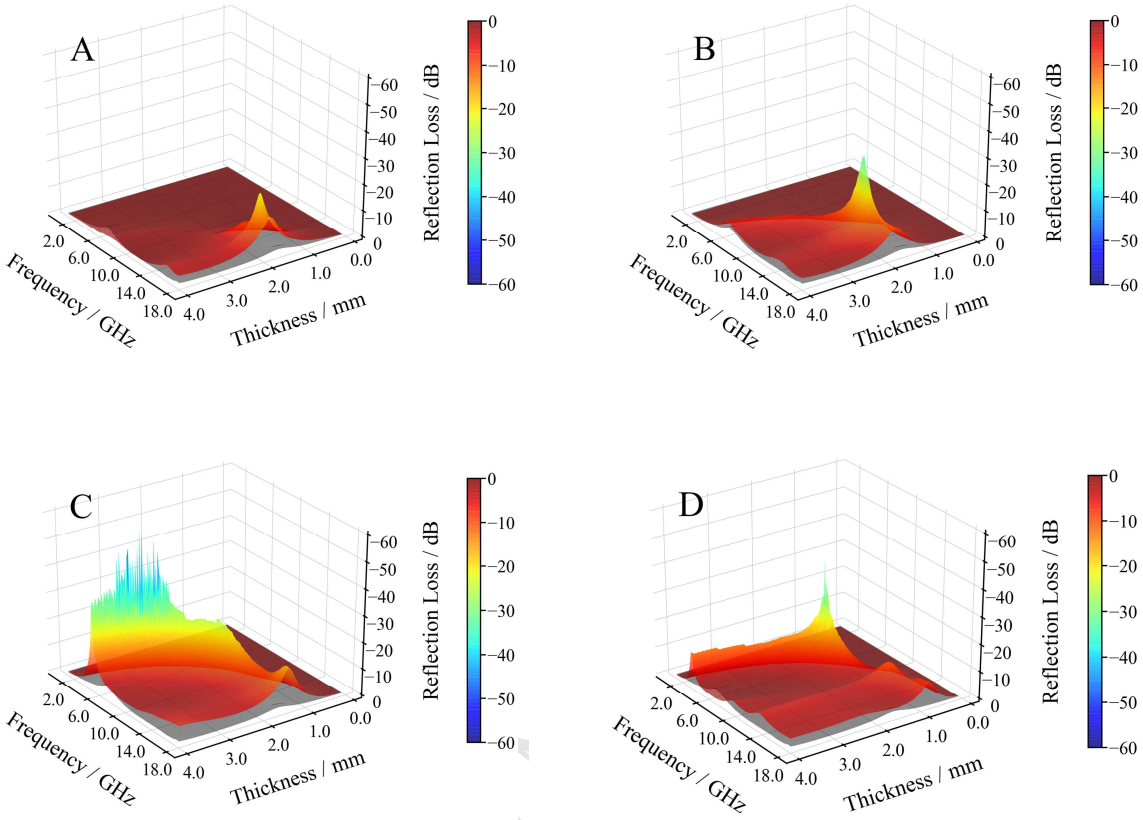


Figure 8. reflection loss domains over the microwave region of 1 – 18 GHz for the hydrogenated TiO_2 nanoparticles treated at (A) 600 °C, (B) 650 °C, (C) 700 °C and (D) 750 °C, overlaid upon the RL domain for anatase (gray).

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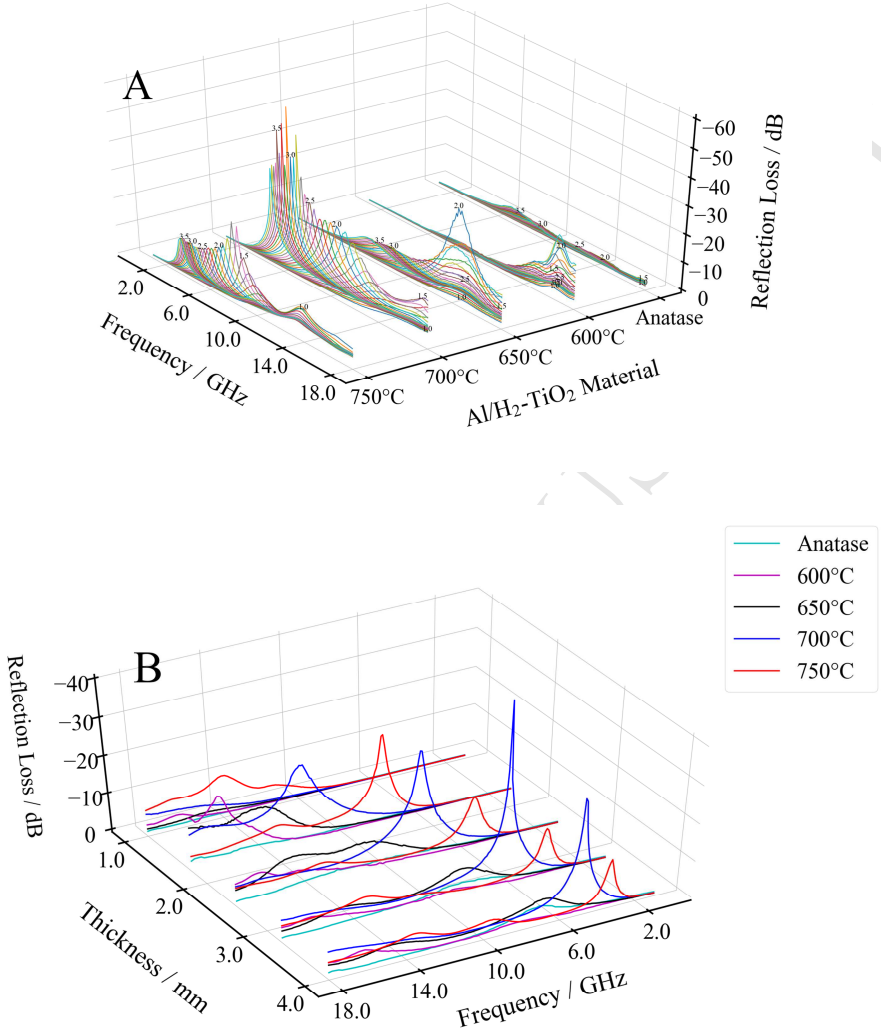


Figure 9. (A) Stacked reflection loss curves over the microwave region of 1 – 18 GHz for both the anatase and hydrogenated TiO_2 nanoparticles treated at 600, 650, 700 and 750 °C. Inserted numerical values are thickness parameters placed directly above the maximal point of their representative curves. (B) Stacked reflection loss curves over the microwave region of 1 – 18 GHz as a function of thickness for anatase (cyan), 600 °C (magenta), 650 °C (black), 700 °C (blue), and 750 °C (red).

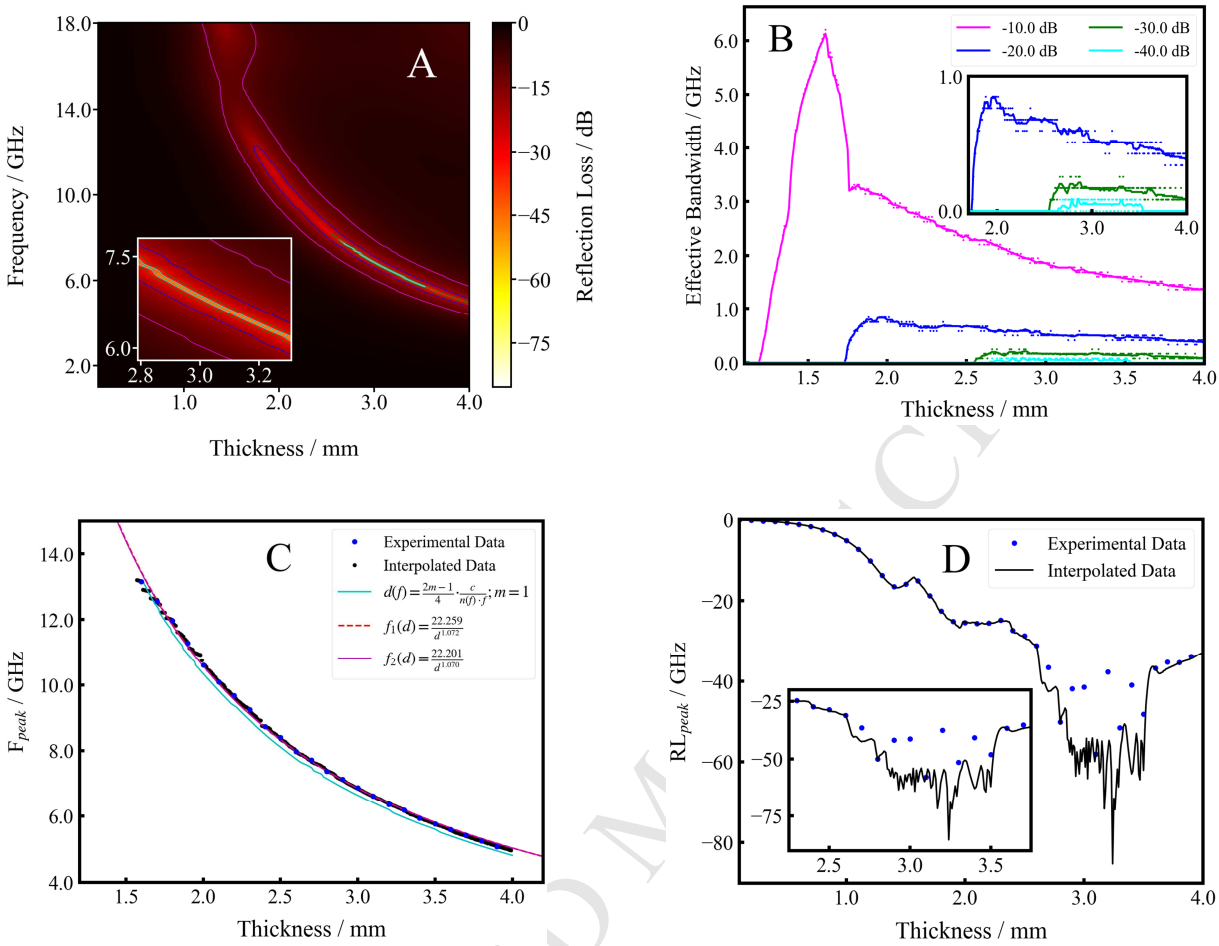


Figure 10. (A) Interpolated Contour Plot of the 700 °C Al/H_2-TiO_2 nanomaterials with demarcated RL thresholds, with (B) the associated effective bandwidth plot (experimental data demarcated as points, interpolated data demarcated by line). (C) The F_{peak} vs. d plot and (D) RL_{peak} vs. d plot from both Experimental (blue demarcations and $f_1(d)$, Δd = 0.1mm, Δf = 0.085GHz)) and Interpolated (black demarcations and $f_2(d)$, Δd = 0.01mm, Δf = 0.01GHz) data sets.

ASSOCIATED CONTENT

Supporting Information. Figures S1-S9.

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

DATA AVAILABILITY

The raw/processed data required to reproduce these findings cannot be shared at this time as the data also forms part of an ongoing study.

CONFLICT OF INTEREST

Authors declare that there are no conflicts of interest.

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SYNOPSIS

Al/H_2 treatment brings enhanced improvement in microwave absorption performance of TiO_2 nanoparticles.

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SUPPLIMENTAL WORK

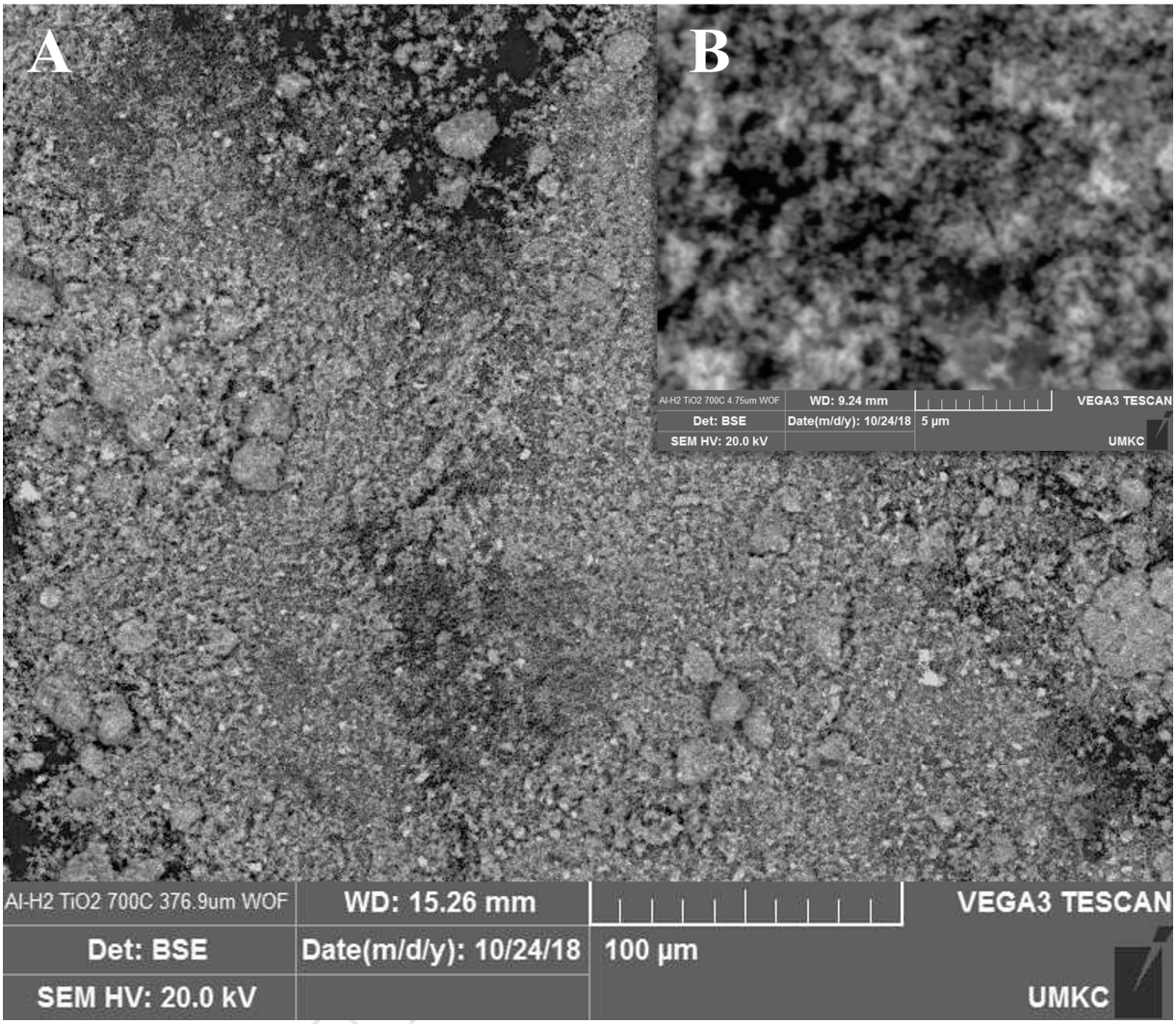


Figure S1: SEM imaging of the Al/H_2-TiO_2 nanoparticles treated at 700 °C at a (A) 376.9µm WOF and (B) 4.75 9µm WOF.

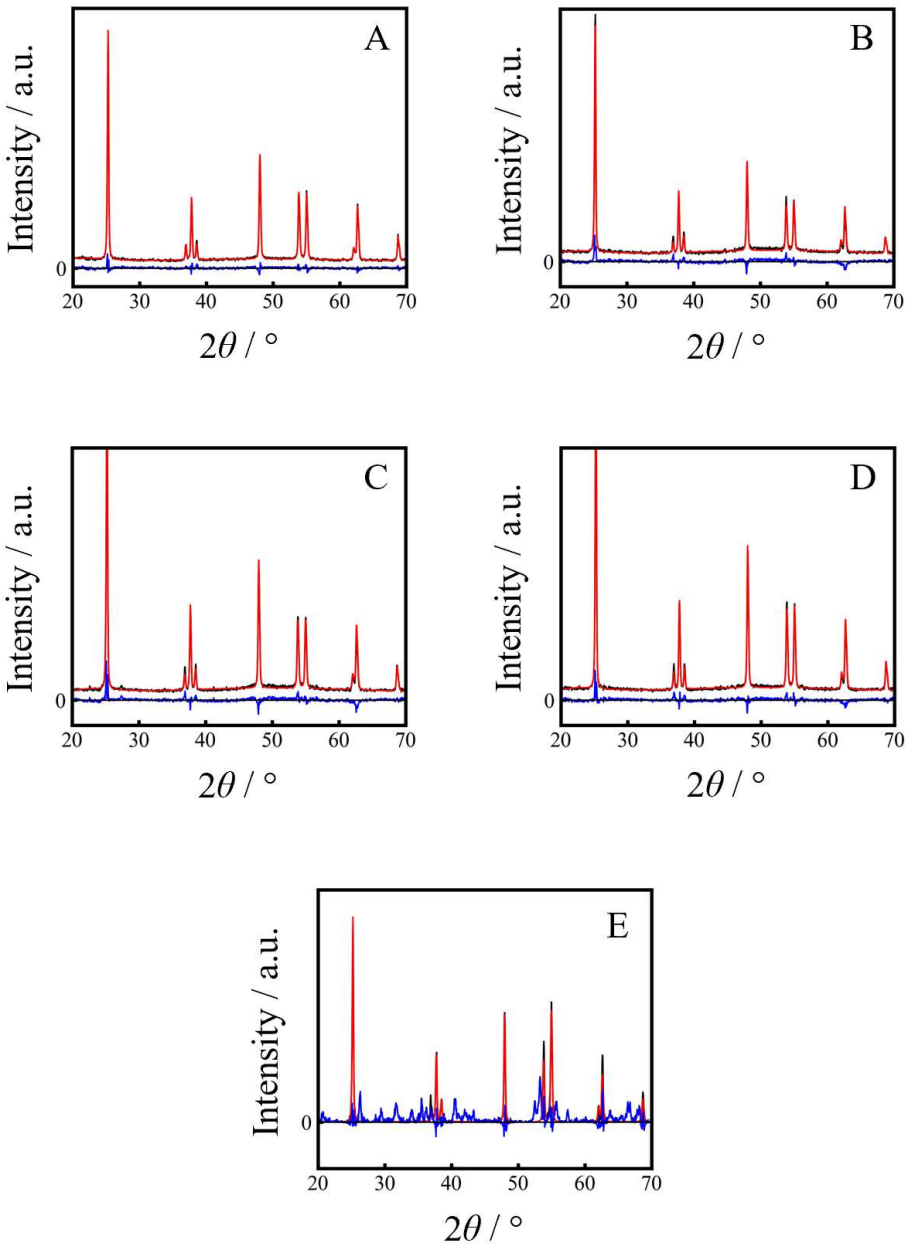
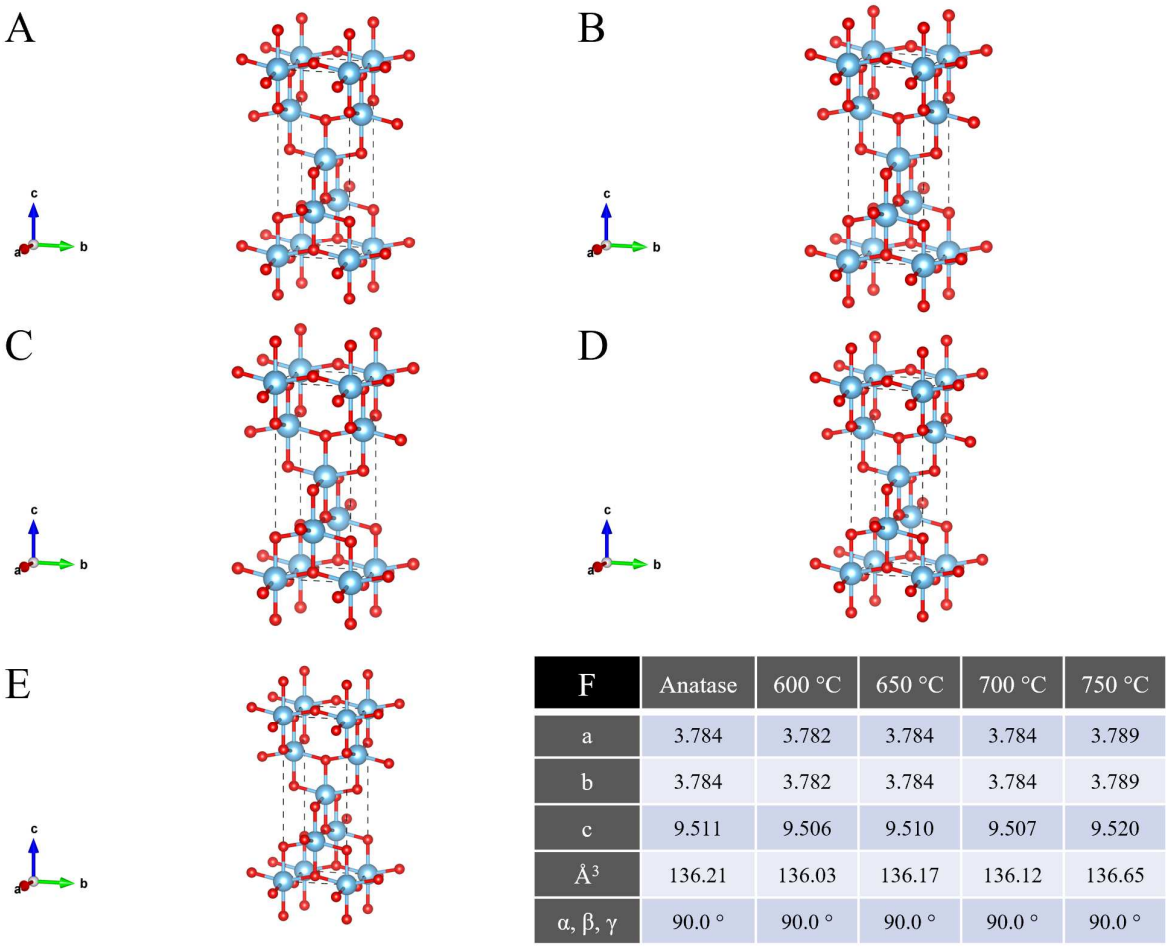


Figure S2: (Black) Observed diffraction pattern, (Red) Rietveld refinement curves, and (Blue) difference curves for the (A) anatase, (B) 600, (C) 650, (D) 700, and (E) 750 °C Al/H_2-TiO_2 nanoparticles.



F	Anatase	600 °C	650 °C	700 °C	750 °C
a	3.784	3.782	3.784	3.784	3.789
b	3.784	3.782	3.784	3.784	3.789
c	9.511	9.506	9.510	9.507	9.520
Å^3	136.21	136.03	136.17	136.12	136.65
α, β, γ	90.0 °	90.0 °	90.0 °	90.0 °	90.0 °

Figure S3: Anatase core crystal resultants from Rietveld refinement for the (A) anatase, (B) 600, (C) 650, (D) 700, and (E) 750 °C Al/H_2-TiO_2 nanoparticles. (F) lattice parameters derived from refinement.

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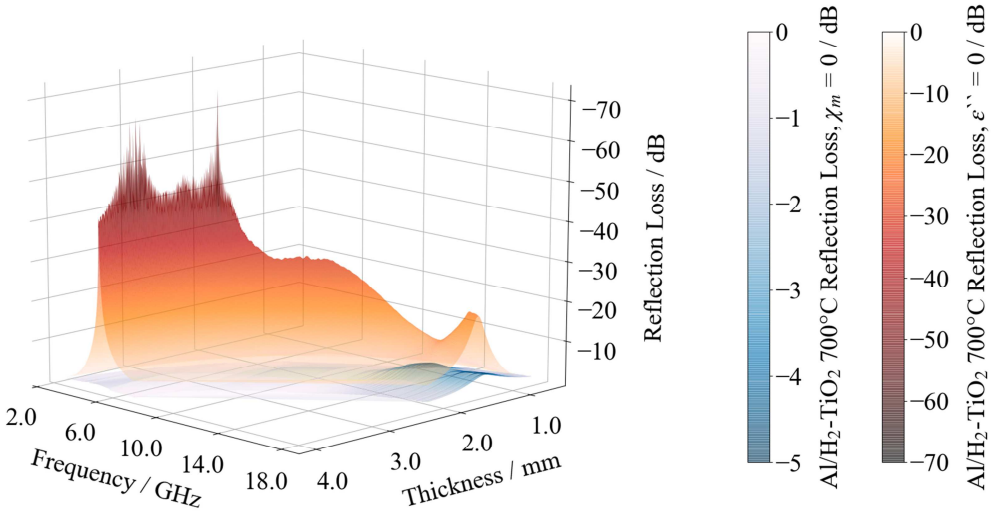


Figure S4: Comparative reflection loss curves for the Al/H_2-TiO_2 material hydrogenated at 700 °C, where the resultant curves were simulated with zero magnetic susceptibility (red) and zero dielectric dissipation (blue).

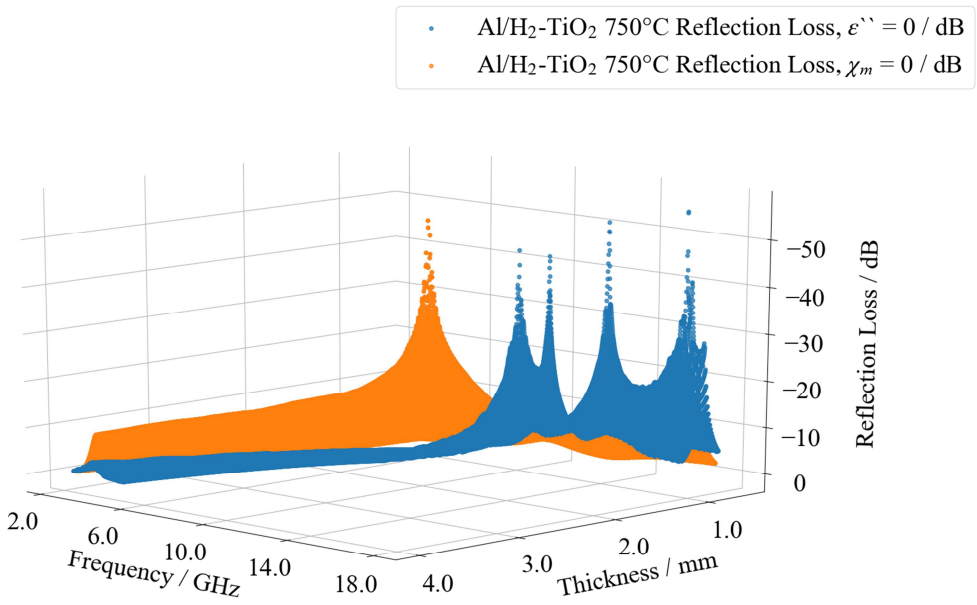


Figure S5: Comparative reflection loss curves for the Al/H_2-TiO_2 material hydrogenated at 750 °C, where the resultant curves were simulated with zero magnetic susceptibility (orange) and zero dielectric dissipation (blue).

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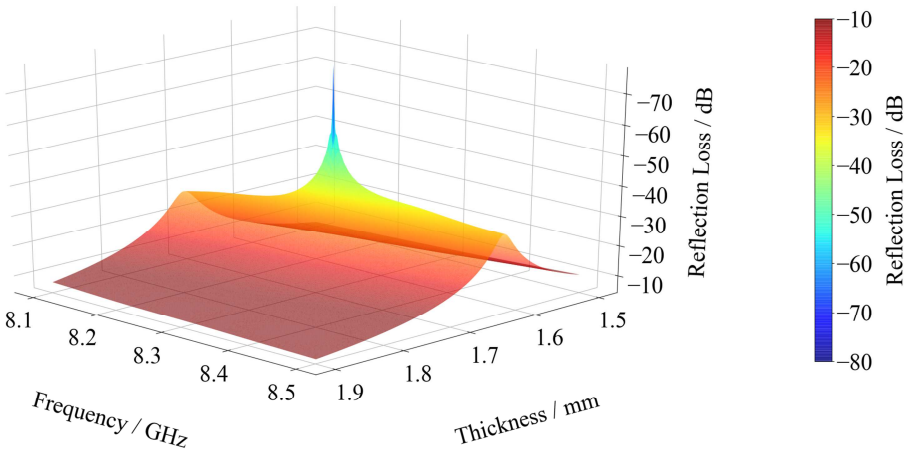


Figure S6. Interpolated plot of the Al/H_2-TiO_2-750 °C peak, using a cubic spline interpolation technique at a 0.01GHz·0.01mm grid space (10^{-2} step size).

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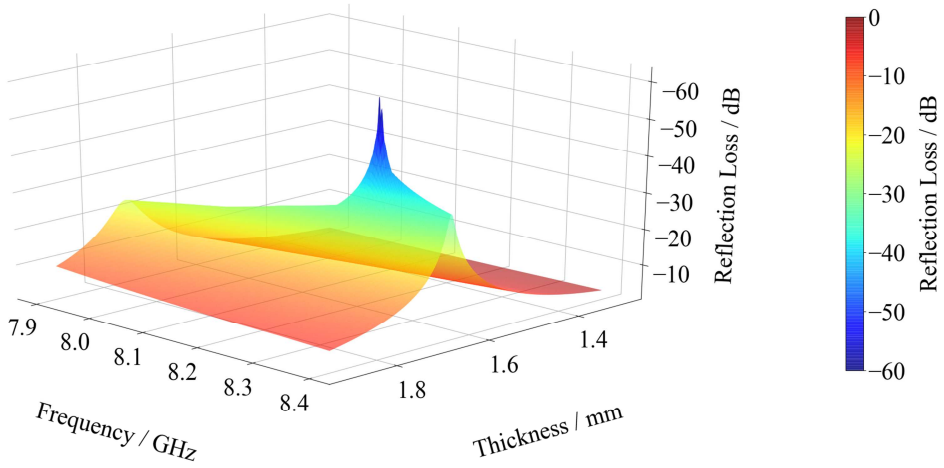


Figure S7. Interpolated plots of Al/H_2-TiO_2-750 °C peak, using a linear interpolation technique over the 0.01GHz·0.01mm gird space (10^{-2} step size).

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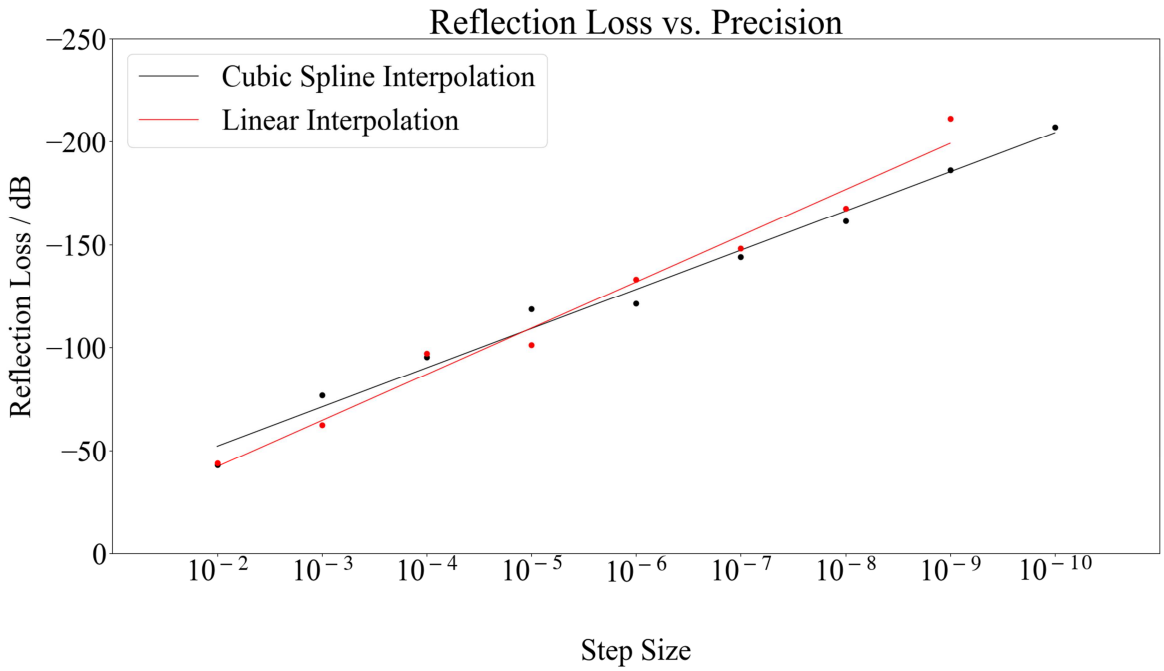


Figure S8. Interpolated reflection loss plots as a function of increasing precision for the Al/H_2-TiO_2-750 °C materials, using cubic and linear spline interpolation techniques.

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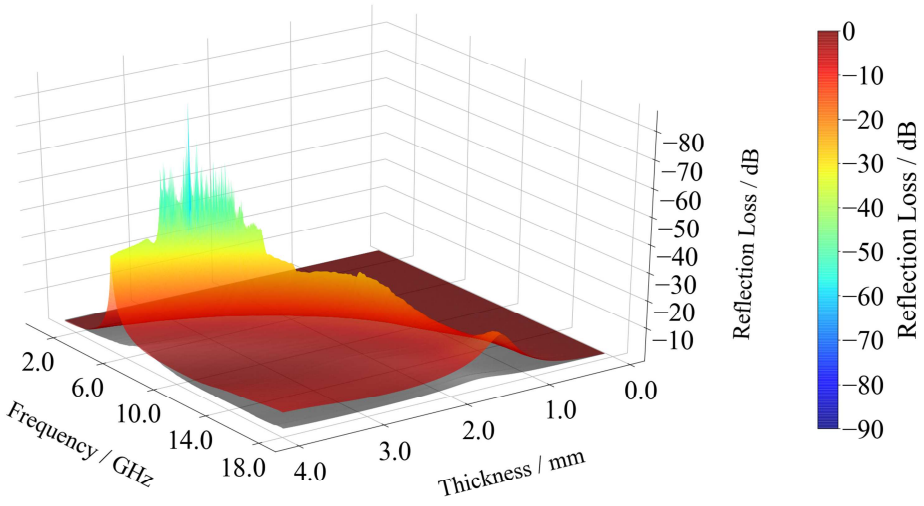


Figure S9. Interpolated plots at 0.01GHz·0.01mm resolution for anatase (gray) and the Al/H_2-TiO_2 nanomaterials hydrogenated at 700 °C (colored).

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BIBLIOGRAPHY

Dr. Xiaobo Chen is an Associate Professor at the University of Missouri–Kansas City, Department of Chemistry. His research interests include nanomaterials, catalysis, electrochemistry, light-materials interactions and their applications in energy, environment and information protection. His renowned work includes the discovery of black TiO_2 with Professor Samuel S. Mao at the University of California, Berkeley and the new application of black TiO_2 nanomaterials along with other nanomaterials in microwave absorption applications. Dr. Chen has published so far 140 peer-reviewed articles with about 37,000 citations.

Mr. Michael Green is a graduate student under the supervision of Dr. Xiaobo Chen at the University of Missouri–Kansas City, Department of Chemistry. He received his Bachelors of Science in chemistry with a minor in mathematics from the University of Idaho in 2016. His research interests include the development, characterization, modeling, and application of nanomaterials in light/matter interactions, focusing on photolysis, photocatalysis, and microwave absorption, as well as short-range matter/matter interactions with a focus in physical adsorption.

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Xiaobo Chen



Michael Green



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Highlights

- Microwave absorption of Al-H_2 treated TiO_2 is reported.
- A large reflection loss (-58.02 dB) has been achieved with 3.1 mm coating
- The reflection loss depends on the treating temperature and its mechanism is discussed.

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